PCF8574 Remote 8-Bit I/O Expander for I²C Bus

1 Features

- Low Standby-Current Consumption of 10 μA Max
- I²C to Parallel-Port Expander
- Open-Drain Interrupt Output
- Compatible With Most Microcontrollers
- Latched Outputs With High-Current Drive Capability for Directly Driving LEDs
- Latch-Up Performance Exceeds 100 mA Per JESD 78, Class II

2 Applications

- Telecom Shelters: Filter Units
- Servers
- Routers (Telecom Switching Equipment)
- Personal Computers
- Personal Electronics
- Industrial Automation
- Products with GPIO-Limited Processors

3 Description

This 8-bit input/output (I/O) expander for the two-line bidirectional bus (I²C) is designed for 2.5-V to 6-V V_{CC} operation.

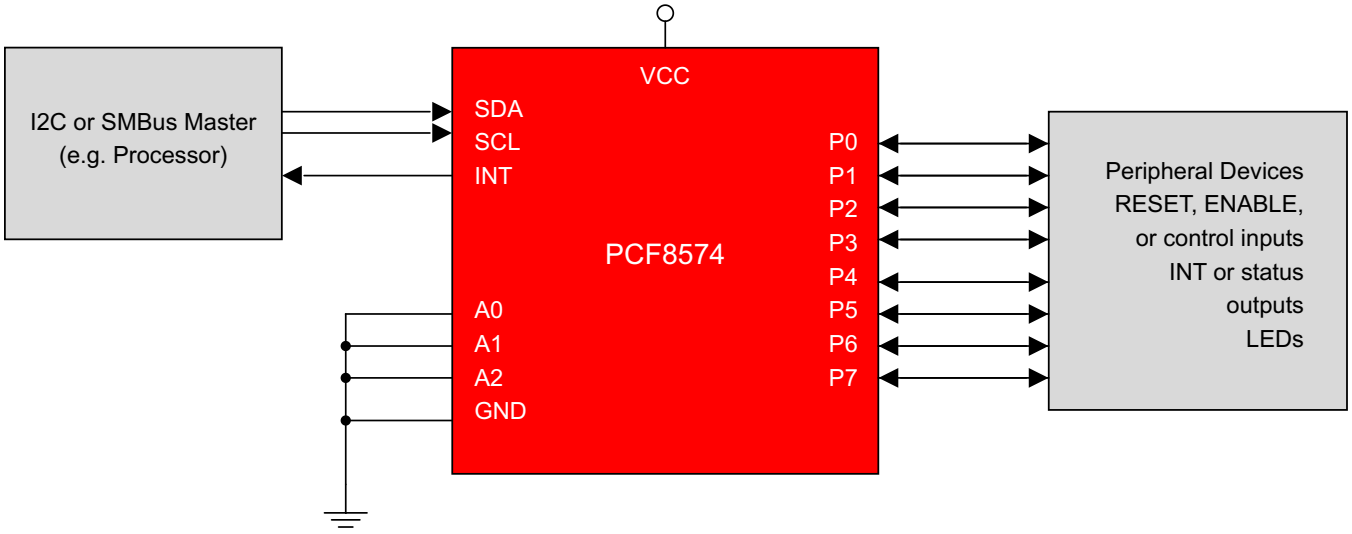
The PCF8574 device provides general-purpose remote I/O expansion for most microcontroller families by way of the I²C interface [serial clock (SCL), serial data (SDA)].

The device features an 8-bit quasi-bidirectional I/O port (P0–P7), including latched outputs with high-current drive capability for directly driving LEDs. Each quasi-bidirectional I/O can be used as an input or output without the use of a data-direction control signal. At power on, the I/Os are high. In this mode, only a current source to V_{CC} is active.

Device Information(1)

PART NUMBER	PACKAGE (PIN)	BODY SIZE (NOM)
PCF8574	TVSOP (20)	5.00 mm × 4.40 mm
	SOIC (16)	10.30 mm × 7.50 mm
	PDIP (16)	19.30 mm × 6.35 mm
	TSSOP (20)	6.50 mm × 4.40 mm
	QFN (16)	3.00 mm × 3.00 mm
	VQFN (20)	4.50 mm × 3.50 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

Changes from Revision I (November 2015) to Revision J — **Page**

- Corrected part number in *Device Information* table ... 1

Changes from Revision H (January 2015) to Revision I — **Page**

- Added Junction temperature to the *Absolute Maximum Ratings* ... 4
- Changed Supply Current (A) To: Supply Current (µA) and f_{SCL} = 400 kHz to f_{SCL} = 100 kHz in Figure 1 ... 6
- Changed Supply Current (A) To: Supply Current (µA) in Figure 1 ... 6
- Changed Supply Current (A) To: Supply Current (µA) and f_{SCL} = 400 kHz to f_{SCL} = 100 kHz in Figure 3 ... 6

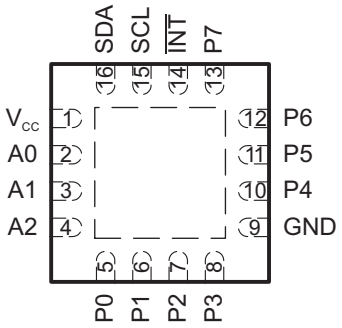
Changes from Revision G (May 2008) to Revision H — **Page**

- Added *Applications*, *Device Information* table, *Pin Functions* table, *ESD Ratings* table, *Thermal Information* table, *Typical Characteristics*, *Feature Description* section, *Device Functional Modes*, *Application and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section. ... 1
- Deleted *Ordering Information* table. ... 1

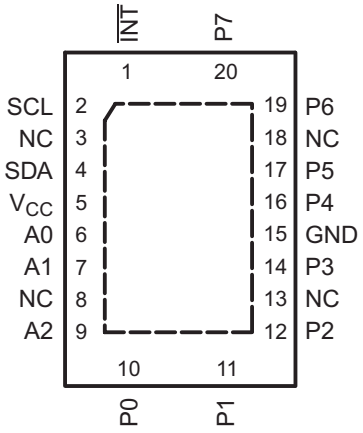
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5 Pin Configuration and Functions

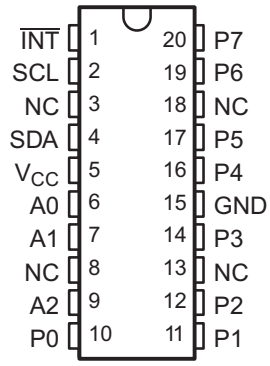
RGT PACKAGE
(TOP VIEW)



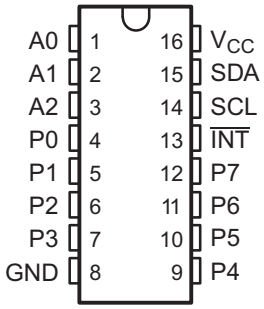
RGY PACKAGE
(TOP VIEW)



DGV OR PW PACKAGE
(TOP VIEW)



DW OR N PACKAGE
(TOP VIEW)



Pin Functions

PIN					TYPE	DESCRIPTION
NAME	RGT	RGY	DGV or PW	DW or N		
A [0..2]	2, 3, 4	6, 7, 9	6, 7, 9	1, 2, 3	I	Address inputs 0 through 2. Connect directly to V_{CC} or ground. Pullup resistors are not needed.
GND	9	15	15	8	—	Ground
$\overline{INT}$	14	1	1	13	O	Interrupt output. Connect to V_{CC} through a pullup resistor.
NC	-	3, 8, 13, 18	3, 8, 13, 18	-	—	Do not connect
P[0..7]	5, 6, 7, 8, 10, 11, 12, 13	10, 11, 12, 14, 16, 17, 19, 20	10, 11, 12, 14, 16, 17, 19, 20	4, 5, 6, 7, 9, 10, 11, 12	I/O	P-port input/output. Push-pull design structure.
SCL	15	2	2	14	I	Serial clock line. Connect to V_{CC} through a pullup resistor
SDA	16	4	4	15	I/O	Serial data line. Connect to V_{CC} through a pullup resistor.
V_{CC}	1	5	5	16	—	Voltage supply

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6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage range		–0.5	7	V
V_I	Input voltage range(2)		–0.5	V_{CC} + 0.5	V
V_O	Output voltage range(2)		–0.5	V_{CC} + 0.5	V
I_{IK}	Input clamp current	V_I < 0		–20	mA
I_{OK}	Output clamp current	V_O < 0		–20	mA
I_{OK}	Input/output clamp current	V_O < 0 or V_O > V_{CC}		±400	μA
I_{OL}	Continuous output low current	V_O = 0 to V_{CC}		50	mA
I_{OH}	Continuous output high current	V_O = 0 to V_{CC}		–4	mA
	Continuous current through V_{CC} or GND			±100	mA
T_J	Junction temperature			150	°C
T_{stg}	Storage temperature range		–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.

6.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	1500	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	2000	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 500-V HBM is possible with the necessary precautions.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 250-V CDM is possible with the necessary precautions.

6.3 Recommended Operating Conditions

		MIN	MAX	UNIT
V_{CC}	Supply voltage	2.5	6	V
V_{IH}	High-level input voltage	0.7 × V_{CC}	V_{CC} + 0.5	V
V_{IL}	Low-level input voltage	–0.5	0.3 × V_{CC}	V
I_{OH}	High-level output current		–1	mA
I_{OL}	Low-level output current		25	mA
T_A	Operating free-air temperature	–40	85	°C

6.4 Thermal Information

THERMAL METRIC(1)		PCF8574						UNIT
		DGV	DW	N	PW	RGT	RGY	
		20 PINS	16 PINS	16 PINS	20 PINS	16 PINS	20 PINS	
θ_{JA}	Junction-to-ambient thermal resistance	92	57	67	83	53	37	°C/W

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report (SPRA953).

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6.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	V_{CC}	MIN	TYP(1)	MAX	UNIT
V_{IK}	Input diode clamp voltage	I_I = –18 mA	2.5 V to 6 V	–1.2			V
V_{POR}	Power-on reset voltage(2)	$V_I = V_{CC}$ or GND, I_O = 0	6 V		1.3	2.4	V
I_{OH}	P port	V_O = GND	2.5 V to 6 V	30		300	µA
I_{OHT}	P port transient pullup current	High during acknowledge, V_{OH} = GND	2.5 V		–1		mA
I_{OL}	SDA	V_O = 0.4 V	2.5 V to 6 V	3			mA
	P port	V_O = 1 V	5 V	10	25		
	$\overline{INT}$	V_O = 0.4 V	2.5 V to 6 V	1.6			
I_I	SCL, SDA	$V_I = V_{CC}$ or GND	2.5 V to 6 V			±5	µA
	$\overline{INT}$					±5	
	A0, A1, A2					±5	
I_{IHL}	P port	$V_I \geq V_{CC}$ or $V_I \leq$ GND	2.5 V to 6 V			±400	µA
I_{CC}	Operating mode	$V_I = V_{CC}$ or GND, I_O = 0, f_{SCL} = 100 kHz	6 V		40	100	µA
	Standby mode	$V_I = V_{CC}$ or GND, I_O = 0			2.5	10	
C_i	SCL	$V_I = V_{CC}$ or GND	2.5 V to 6 V		1.5	7	pF
C_{io}	SDA	$V_{IO} = V_{CC}$ or GND	2.5 V to 6 V		3	7	pF
	P port				4	10	

(1) All typical values are at V_{CC} = 5 V, T_A = 25°C.
(2) The power-on reset circuit resets the I^2C-bus logic with $V_{CC} < V_{POR}$ and sets all I/Os to logic high (with current source to V_{CC}).

6.6 I^2C Interface Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 12)

			MIN	MAX	UNIT
f_{scl}	I^2C clock frequency			100	kHz
t_{sch}	I^2C clock high time		4		µs
t_{scl}	I^2C clock low time		4.7		µs
t_{sp}	I^2C spike time			100	ns
t_{sds}	I^2C serial data setup time		250		ns
t_{sdh}	I^2C serial data hold time		0		ns
t_{icr}	I^2C input rise time			1	µs
t_{icf}	I^2C input fall time			0.3	µs
t_{ocf}	I^2C output fall time (10-pF to 400-pF bus)			300	ns
t_{buf}	I^2C bus free time between stop and start		4.7		µs
t_{sts}	I^2C start or repeated start condition setup		4.7		µs
t_{sth}	I^2C start or repeated start condition hold		4		µs
t_{sps}	I^2C stop condition setup		4		µs
t_{vd}	Valid data time	SCL low to SDA output valid		3.4	µs
C_b	I^2C bus capacitive load			400	pF

6.7 Switching Characteristics

over recommended operating free-air temperature range, $C_L \leq$ 100 pF (unless otherwise noted) (see Figure 13)

PARAMETER		FROM (INPUT)	TO (OUTPUT)	MIN	MAX	UNIT
t_{pv}	Output data valid	SCL	P port		4	µs
t_{su}	Input data setup time	P port	SCL	0		µs
t_h	Input data hold time	P port	SCL	4		µs
t_{iv}	Interrupt valid time	P port	$\overline{INT}$		4	µs
t_{ir}	Interrupt reset delay time	SCL	$\overline{INT}$		4	µs

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6.8 Typical Characteristics

T_A = 25°C (unless otherwise noted)

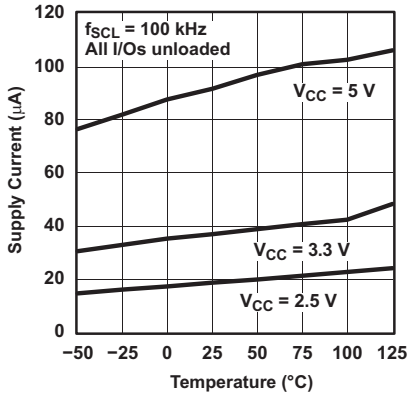


Figure 1. Supply Current vs Temperature

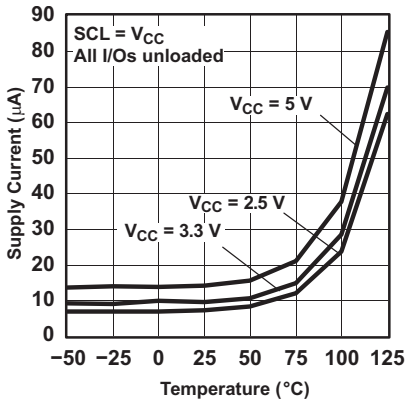


Figure 2. Standby Supply Current vs Temperature

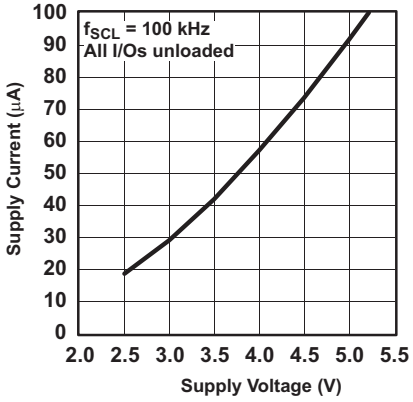


Figure 3. Supply Current vs Supply Voltage

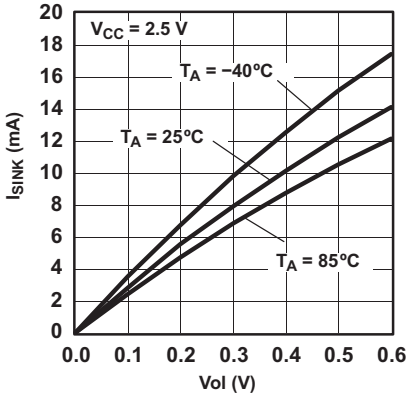


Figure 4. I/O Sink Current vs Output Low Voltage

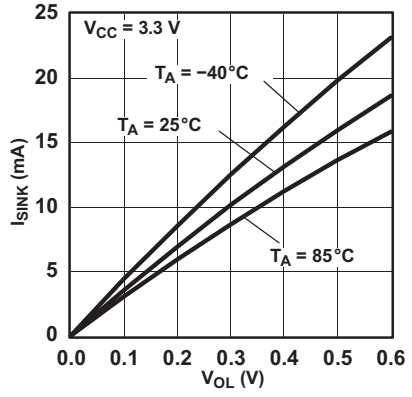


Figure 5. I/O Sink Current vs Output Low Voltage

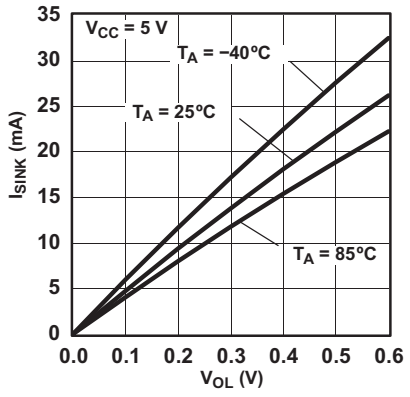


Figure 6. I/O Sink Current vs Output Low Voltage

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Typical Characteristics (continued)

T_A = 25°C (unless otherwise noted)

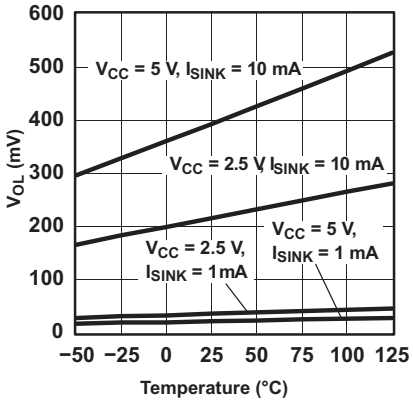


Figure 7. I/O Output Low Voltage vs Temperature

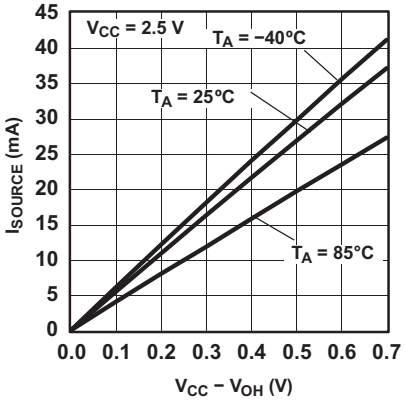


Figure 8. I/O Source Current vs Output High Voltage

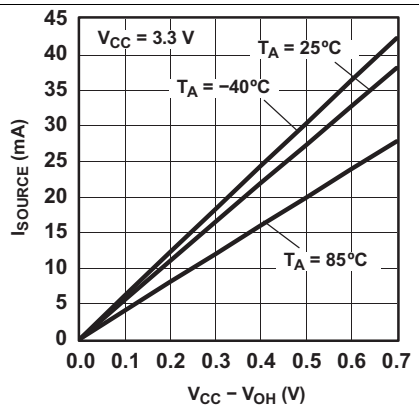


Figure 9. I/O Source Current vs Output High Voltage

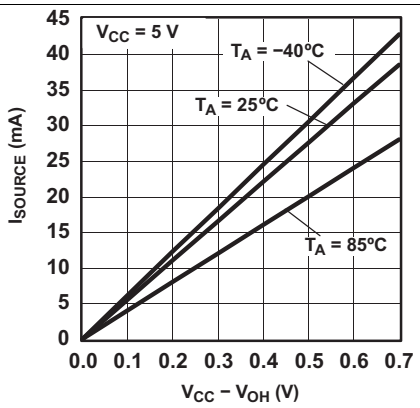


Figure 10. I/O Source Current vs Output High Voltage

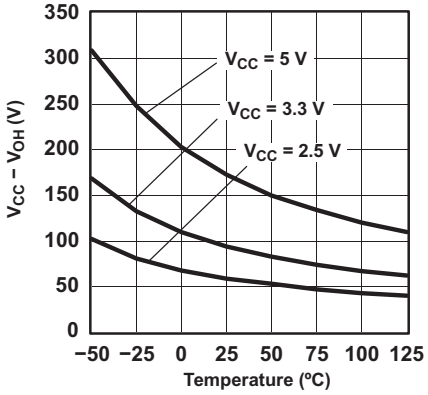


Figure 11. I/O High Voltage vs Temperature

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7 Parameter Measurement Information

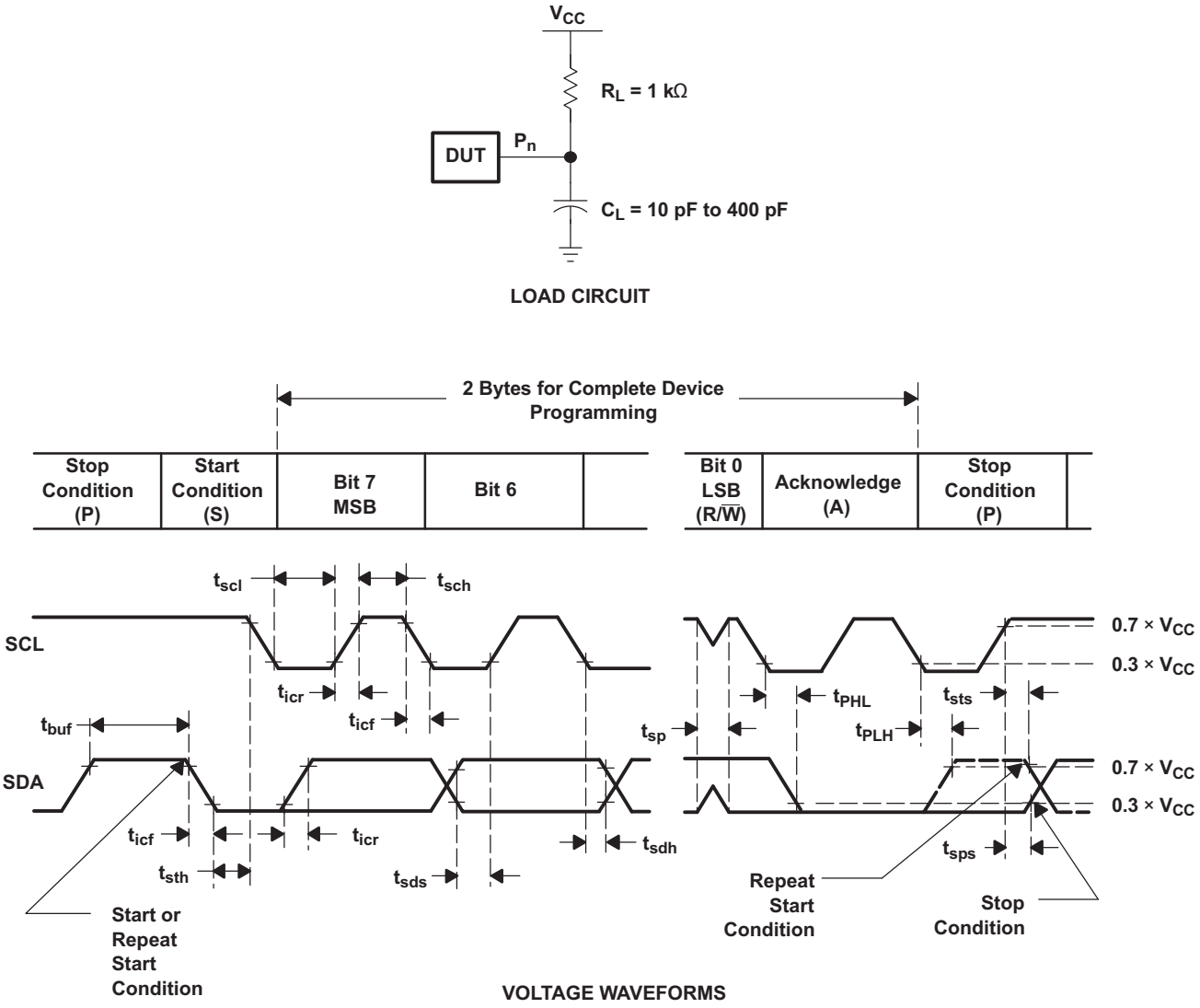


Figure 12. I²C Interface Load Circuit and Voltage Waveforms

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Parameter Measurement Information (continued)

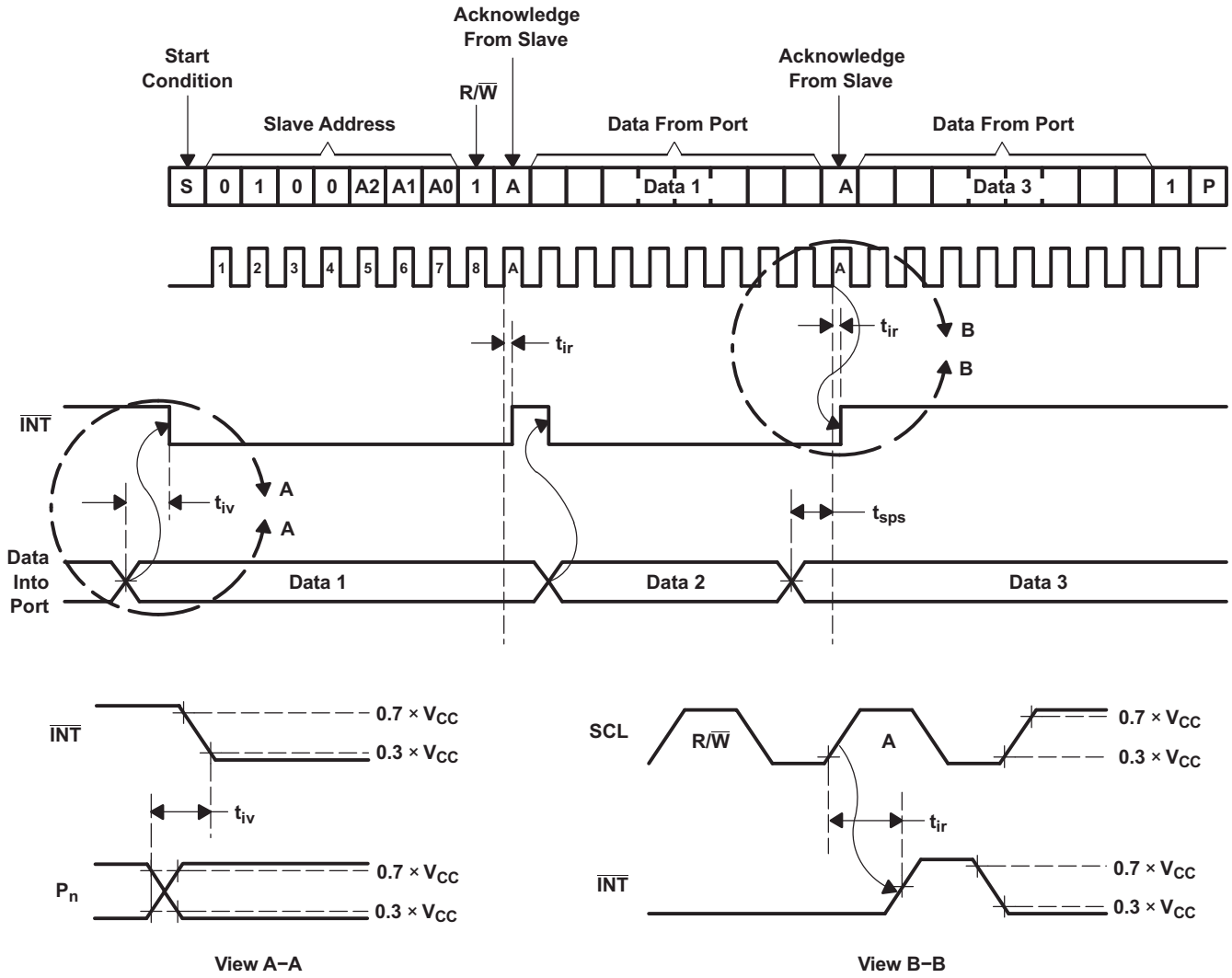


Figure 13. Interrupt Voltage Waveforms

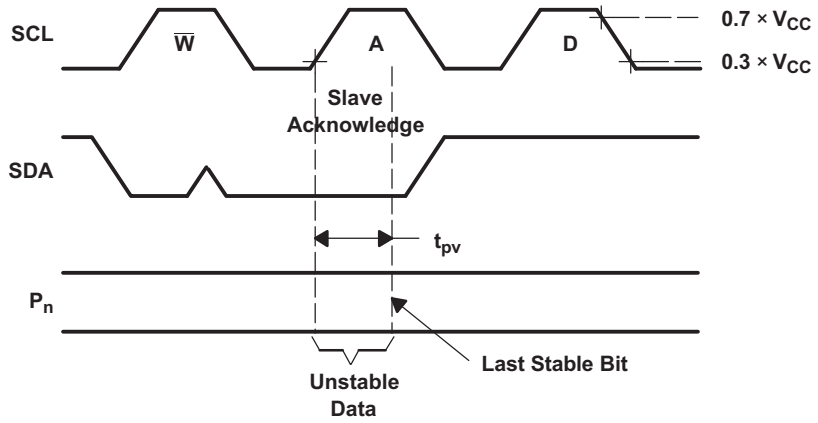


Figure 14. I²C Write Voltage Waveforms

Parameter Measurement Information (continued)

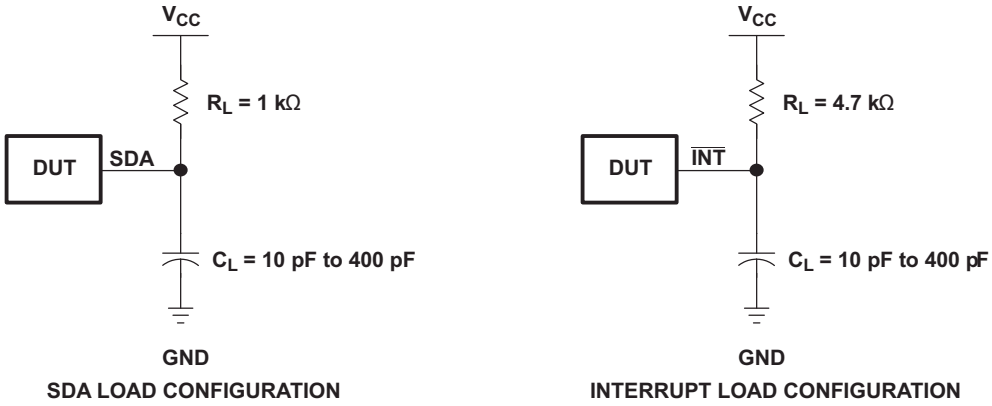


Figure 15. Load Circuits

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8 Detailed Description

8.1 Overview

The PCF8574 device is an 8-bit I/O expander for the two-line bidirectional bus (I2C) is designed for 2.5-V to 5.5-V V_{CC} operation. It provides general-purpose remote I/O expansion for most micro-controller families via the I2C interface (serial clock, SCL, and serial data, SDA, pins).

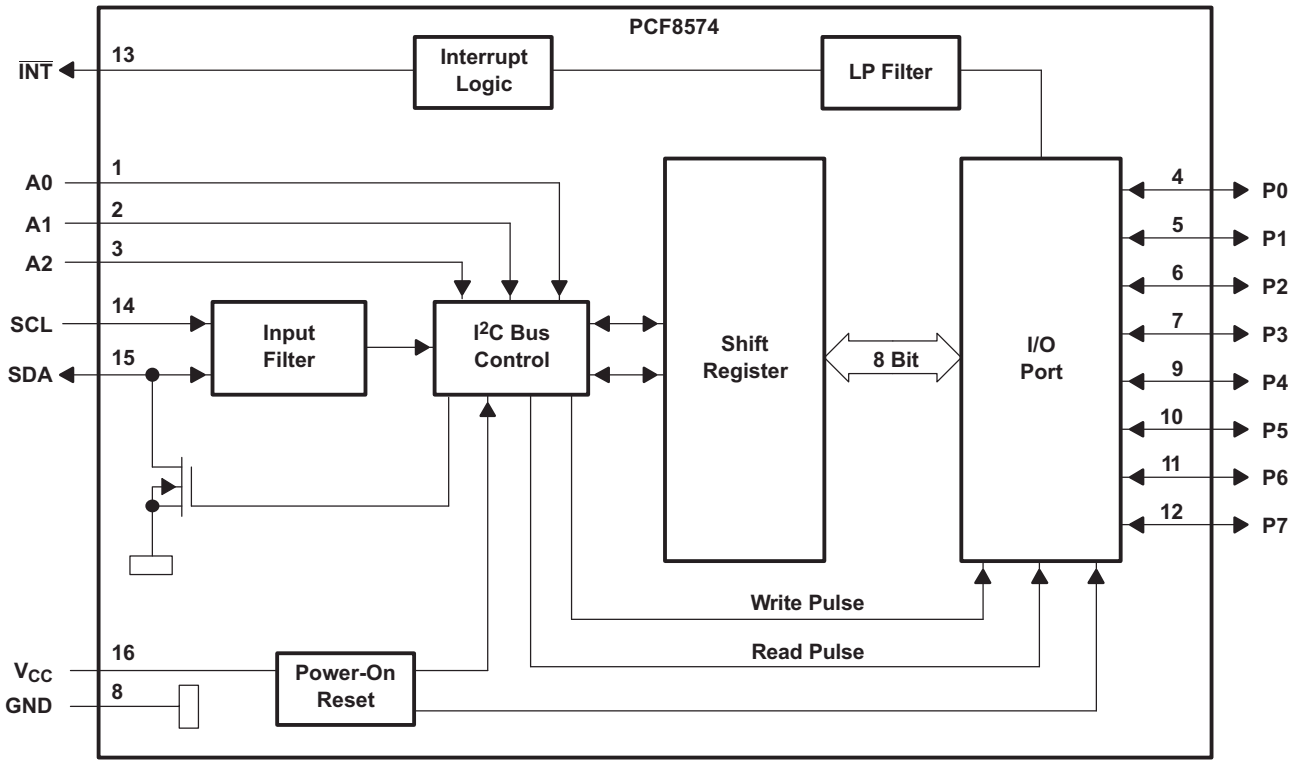
The PCF8574 device provides an open-drain output ($\overline{INT}$) that can be connected to the interrupt input of a microcontroller. An interrupt is generated by any rising or falling edge of the port inputs in the input mode. After time, t_{iv}, $\overline{INT}$ is valid. Resetting and reactivating the interrupt circuit is achieved when data on the port is changed to the original setting or data is read from, or written to, the port that generated the interrupt. Resetting occurs in the read mode at the acknowledge bit after the rising edge of the SCL signal, or in the write mode at the acknowledge bit after the high-to-low transition of the SCL signal. Interrupts that occur during the acknowledge clock pulse can be lost (or be very short) due to the resetting of the interrupt during this pulse. Each change of the I/Os after resetting is detected and, after the next rising clock edge, is transmitted as $\overline{INT}$. Reading from, or writing to, another device does not affect the interrupt circuit. This device does not have internal configuration or status registers. Instead, read or write to the device I/Os directly after sending the device address (see Figure 16 and Figure 17).

By sending an interrupt signal on this line, the remote I/O can inform the microcontroller if there is incoming data on its ports without having to communicate by way of the I^2C bus. Therefore, PCF8574 can remain a simple slave device.

An additional strong pullup to V_{CC} allows fast rising edges into heavily loaded outputs. This device turns on when an output is written high and is switched off by the negative edge of SCL. The I/Os should be high before being used as inputs.

8.2 Functional Block Diagram

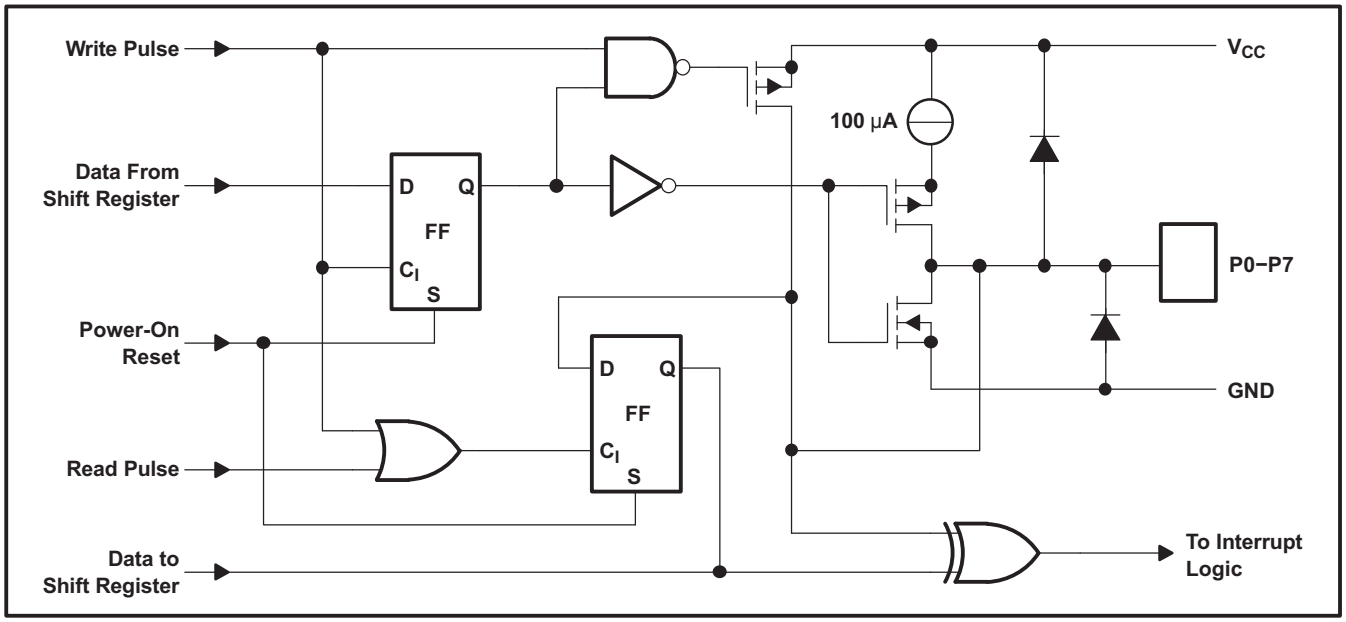
8.2.1 Simplified Block Diagram of Device



Pin numbers shown are for the DW and N packages.

Functional Block Diagram (continued)

8.2.2 Simplified Schematic Diagram of Each P-Port Input/Output



8.3 Feature Description

8.3.1 I²C Interface

I²C communication with this device is initiated by a master sending a start condition, a high-to-low transition on the SDA I/O while the SCL input is high. After the start condition, the device address byte is sent, most-significant bit (MSB) first, including the data direction bit (R/$\overline{W}$). This device does not respond to the general call address. After receiving the valid address byte, this device responds with an acknowledge, a low on the SDA I/O during the high of the acknowledge-related clock pulse. The address inputs (A0–A2) of the slave device must not be changed between the start and the stop conditions.

The data byte follows the address acknowledge. If the R/$\overline{W}$ bit is high, the data from this device are the values read from the P port. If the R/$\overline{W}$ bit is low, the data are from the master, to be output to the P port. The data byte is followed by an acknowledge sent from this device. If other data bytes are sent from the master, following the acknowledge, they are ignored by this device. Data are output only if complete bytes are received and acknowledged. The output data will be valid at time, t_{pv}, after the low-to-high transition of SCL and during the clock cycle for the acknowledge.

A stop condition, which is a low-to-high transition on the SDA I/O while the SCL input is high, is sent by the master.

8.3.2 Interface Definition

BYTE	BIT							
	7 (MSB)	6	5	4	3	2	1	0 (LSB)
I²C slave address	L	H	L	L	A2	A1	A0	R/$\overline{W}$
I/O data bus	P7	P6	P5	P4	P3	P2	P1	P0

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8.3.3 Address Reference

INPUTS			I^2C BUS SLAVE 8-BIT READ ADDRESS	I^2C BUS SLAVE 8-BIT WRITE ADDRESS
A2	A1	A0		
L	L	L	65 (decimal), 41 (hexadecimal)	64 (decimal), 40 (hexadecimal)
L	L	H	67 (decimal), 43 (hexadecimal)	66 (decimal), 42 (hexadecimal)
L	H	L	69 (decimal), 45 (hexadecimal)	68 (decimal), 44 (hexadecimal)
L	H	H	71 (decimal), 47 (hexadecimal)	70 (decimal), 46 (hexadecimal)
H	L	L	73 (decimal), 49 (hexadecimal)	72 (decimal), 48 (hexadecimal)
H	L	H	75 (decimal), 4B (hexadecimal)	74 (decimal), 4A (hexadecimal)
H	H	L	77 (decimal), 4D (hexadecimal)	76 (decimal), 4C (hexadecimal)
H	H	H	79 (decimal), 4F (hexadecimal)	78 (decimal), 4E (hexadecimal)

8.4 Device Functional Modes

Figure 16 and Figure 17 show the address and timing diagrams for the write and read modes, respectively.

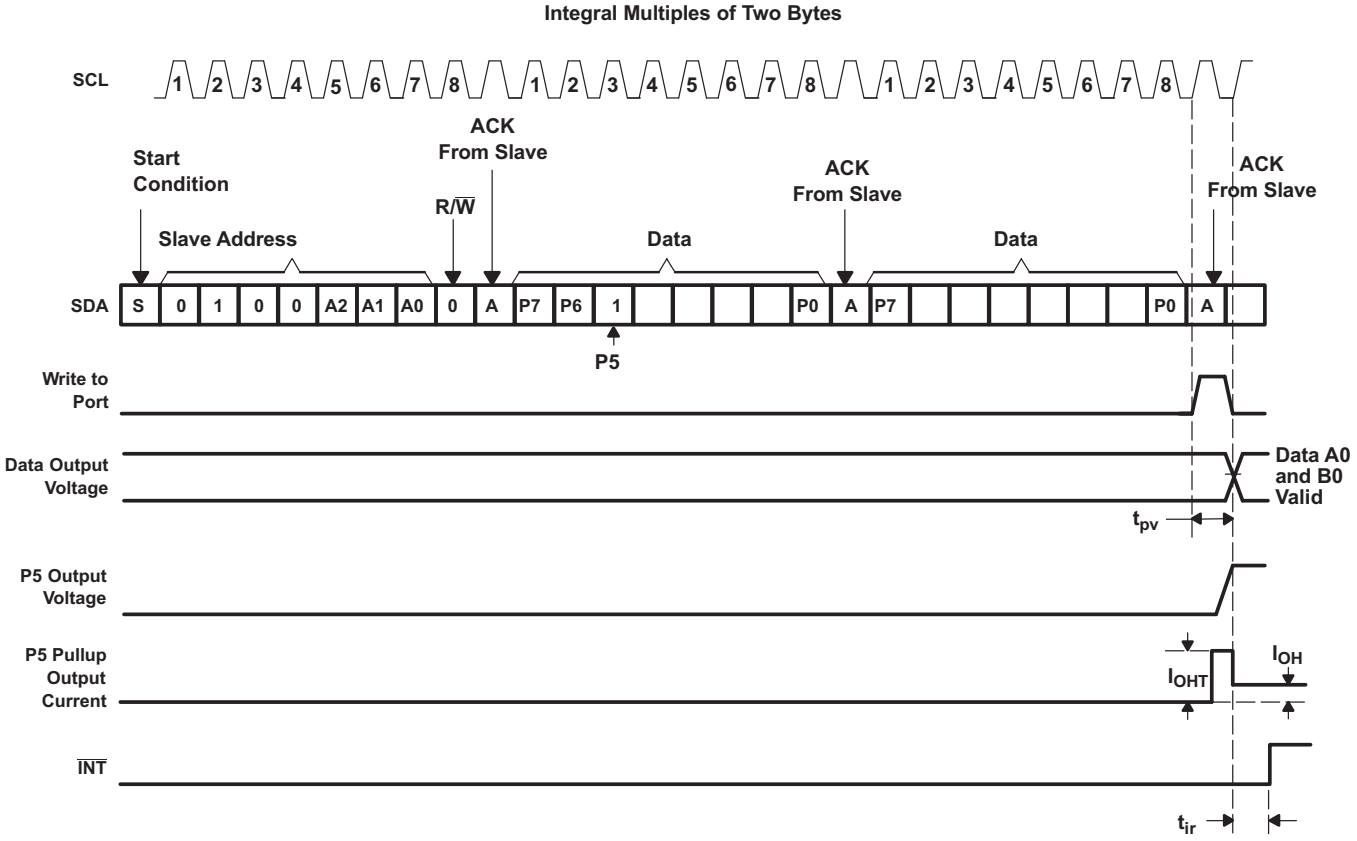
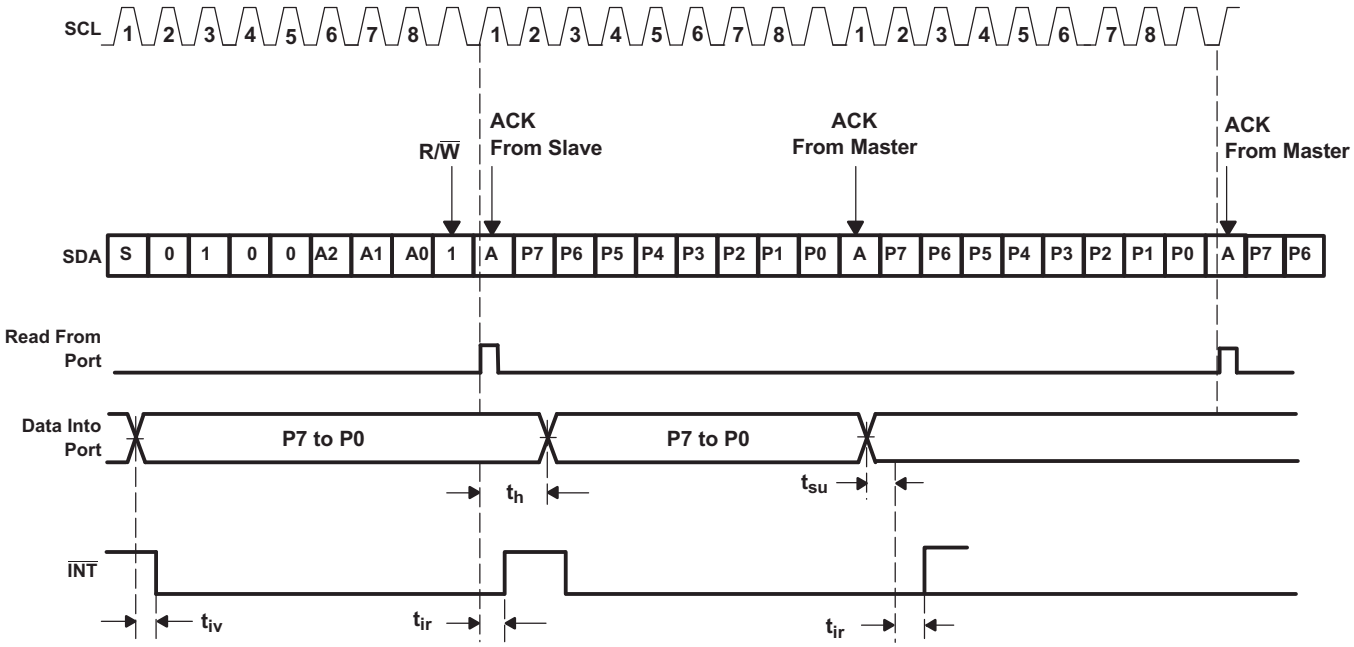


Figure 16. Write Mode (Output)

Device Functional Modes (continued)



A. A low-to-high transition of SDA while SCL is high is defined as the stop condition (P). The transfer of data can be stopped at any moment bya stop condition. When this occurs, data present at the latest ACK phase is valid (output mode). Input data is lost.

Figure 17. Read Mode (Input)

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9 Application and Implementation

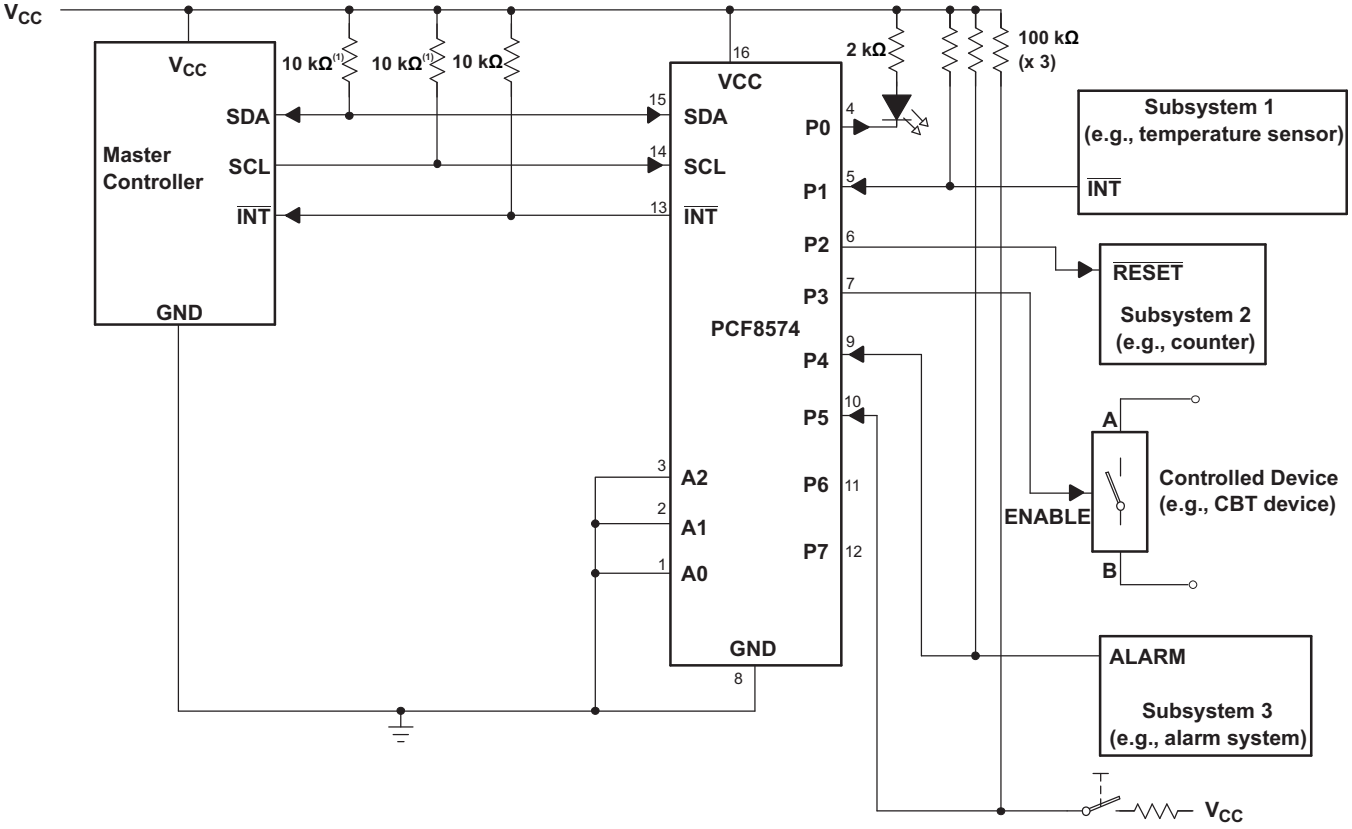
NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

Figure 18 shows an application in which the PCF8574 device can be used.

9.2 Typical Application



(1) The SCL and SDA pins must be tied directly to V_{CC} because if SCL and SDA are tied to an auxiliary power supply that could be powered on while V_{CC} is powered off, then the supply current, ICC, will increase as a result.

A. Device address is configured as 0100000 for this example.

B. P0, P2, and P3 are configured as outputs.

C. P1, P4, and P5 are configured as inputs.

D. P6 and P7 are not used and must be configured as outputs.

Figure 18. Application Schematic

Typical Application (continued)

9.2.1 Design Requirements

9.2.1.1 Minimizing I_{CC} When I/Os Control LEDs

When the I/Os are used to control LEDs, normally they are connected to V_{CC} through a resistor as shown in Figure 18. For a P-port configured as an input, I_{CC} increases as V_I becomes lower than V_{CC}. The LED is a diode, with threshold voltage V_T, and when a P-port is configured as an input the LED will be off but V_I is a V_T drop below V_{CC}.

For battery-powered applications, it is essential that the voltage of P-ports controlling LEDs is greater than or equal to V_{CC} when the P-ports are configured as input to minimize current consumption. Figure 19 shows a high-value resistor in parallel with the LED. Figure 20 shows V_{CC} less than the LED supply voltage by at least V_T. Both of these methods maintain the I/O V_I at or above V_{CC} and prevents additional supply current consumption when the P-port is configured as an input and the LED is off.

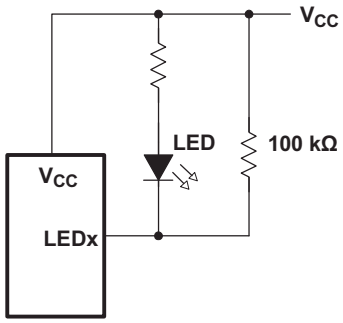


Figure 19. High-Value Resistor in Parallel With LED

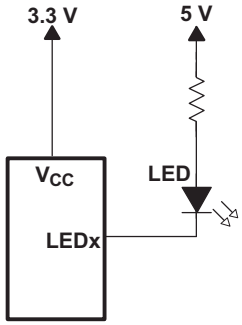


Figure 20. Device Supplied by a Lower Voltage

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Typical Application (continued)

9.2.2 Detailed Design Procedure

The pull-up resistors, R_P, for the SCL and SDA lines need to be selected appropriately and take into consideration the total capacitance of all slaves on the I^2C bus. The minimum pull-up resistance is a function of V_{CC}, $V_{OL,(max)}$, and I_{OL}:

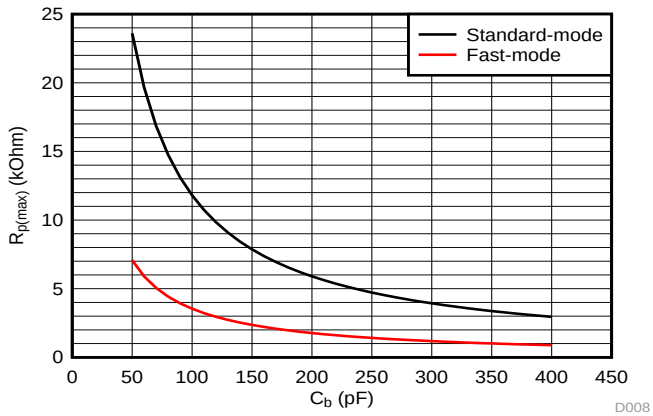
$$R_{p(min)} = \frac{V_{CC} - V_{OL(max)}}{I_{OL}} \quad (1)$$

The maximum pull-up resistance is a function of the maximum rise time, t_r (300 ns for fast-mode operation, f_{SCL} = 400 kHz) and bus capacitance, C_b:

$$R_{p(max)} = \frac{t_r}{0.8473 \times C_b} \quad (2)$$

The maximum bus capacitance for an I^2C bus must not exceed 400 pF for standard-mode or fast-mode operation. The bus capacitance can be approximated by adding the capacitance of the PCF8574 device, C_i for SCL or C_{io} for SDA, the capacitance of wires/connections/traces, and the capacitance of additional slaves on the bus.

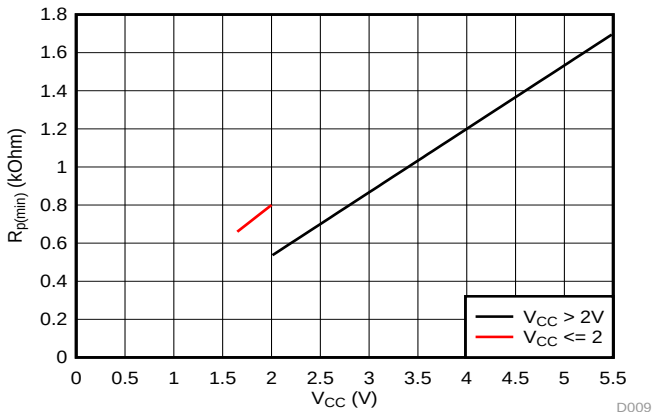
9.2.3 Application Curves



Standard-mode (f_{SCL}= 100 kHz, t_r = 1 µs)

Fast-mode (f_{SCL}= 400 kHz, t_r= 300 ns)

Figure 21. Maximum Pull-Up resistance ($R_{p(max)}$) vs Bus Capacitance (C_b)



V_{OL} = 0.2*V_{CC}, I_{OL} = 2 mA when V_{CC} ≤ 2 V

V_{OL} = 0.4 V, I_{OL} = 3 mA when V_{CC} > 2 V

Figure 22. Minimum Pull-Up Resistance ($R_{p(min)}$) vs Pull-Up Reference Voltage (V_{CC})

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10 Power Supply Recommendations

10.1 Power-On Reset Requirements

In the event of a glitch or data corruption, the PCF8574 device can be reset to its default conditions by using the power-on reset feature. Power-on reset requires that the device go through a power cycle to be completely reset. This reset also happens when the device is powered on for the first time in an application.

The two types of power-on reset are shown in Figure 23 and Figure 24.

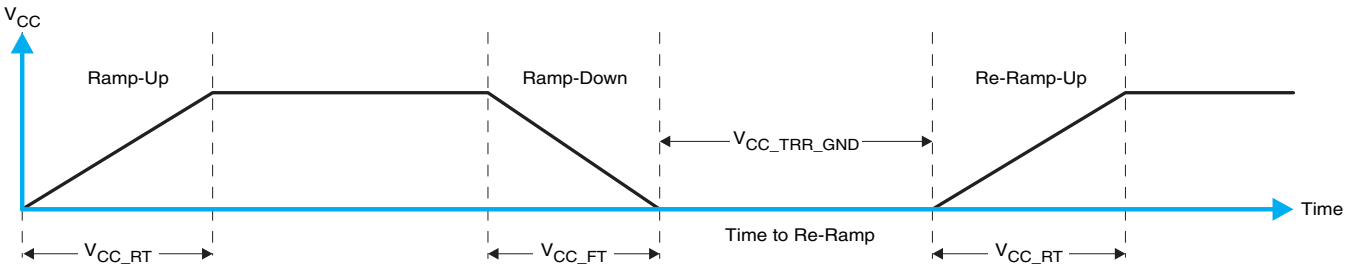


Figure 23. V_{CC} is Lowered Below 0.2 V or 0 V and Then Ramped Up to V_{CC}

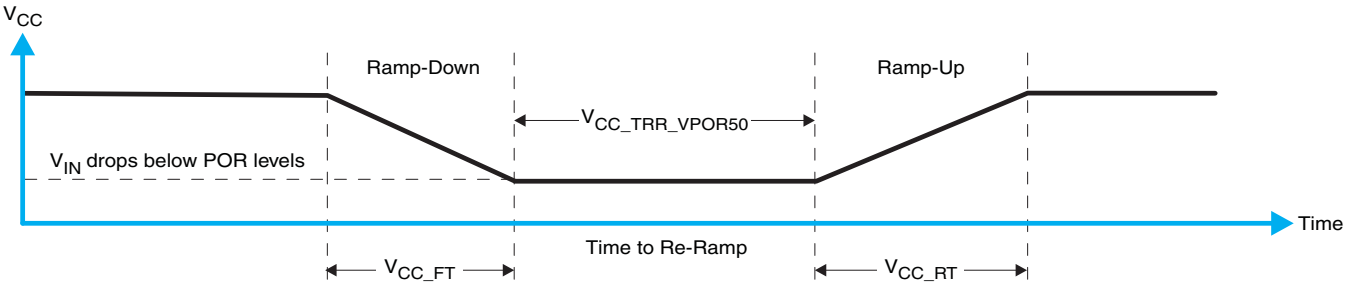


Figure 24. V_{CC} is Lowered Below the POR Threshold, Then Ramped Back Up to V_{CC}

Table 1 specifies the performance of the power-on reset feature for PCF8574 for both types of power-on reset.

Table 1. Recommended Supply Sequencing and Ramp Rates(1)

PARAMETER			MIN	TYP	MAX	UNIT
V_{CC_FT}	Fall rate	See Figure 23	1		100	ms
V_{CC_RT}	Rise rate	See Figure 23	0.01		100	ms
$V_{CC_TRR_GND}$	Time to re-ramp (when V_{CC} drops to GND)	See Figure 23	0.001			ms
$V_{CC_TRR_POR50}$	Time to re-ramp (when V_{CC} drops to V_{POR_MIN} – 50 mV)	See Figure 24	0.001			ms
V_{CC_GH}	Level that V_{CCP} can glitch down to, but not cause a functional disruption when V_{CCX_GW} = 1 µs	See Figure 25			1.2	V
V_{CC_GW}	Glitch width that will not cause a functional disruption when V_{CCX_GH} = 0.5 × V_{CCx}	See Figure 25				µs
V_{PORF}	Voltage trip point of POR on falling V_{CC}		0.767		1.144	V
V_{PORR}	Voltage trip point of POR on fising V_{CC}		1.033		1.428	V

(1) T_A = –40°C to 85°C (unless otherwise noted)

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Glitches in the power supply can also affect the power-on reset performance of this device. The glitch width (V_{CC_GW}) and height (V_{CC_GH}) are dependent on each other. The bypass capacitance, source impedance, and device impedance are factors that affect power-on reset performance. Figure 25 and Table 1 provide more information on how to measure these specifications.

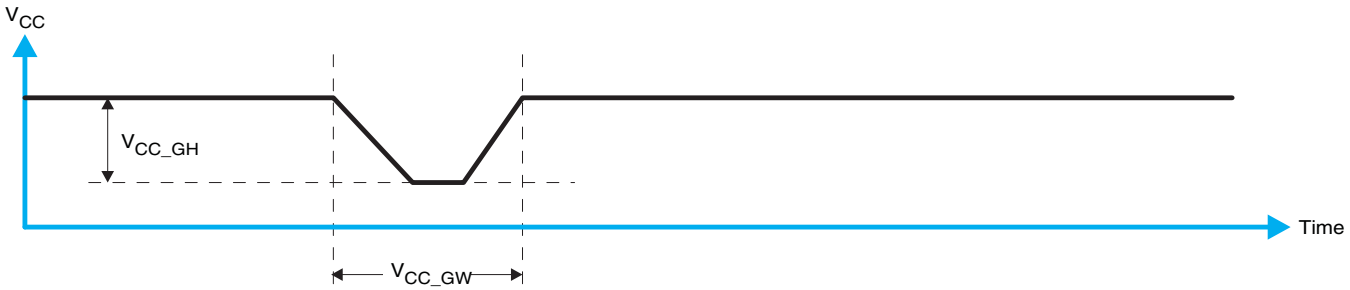


Figure 25. Glitch Width and Glitch Height

V_{POR} is critical to the power-on reset. V_{POR} is the voltage level at which the reset condition is released and all the registers and the I^2C/SMBus state machine are initialized to their default states. The value of V_{POR} differs based on the V_{CC} being lowered to or from 0. Figure 26 and Table 1 provide more details on this specification.

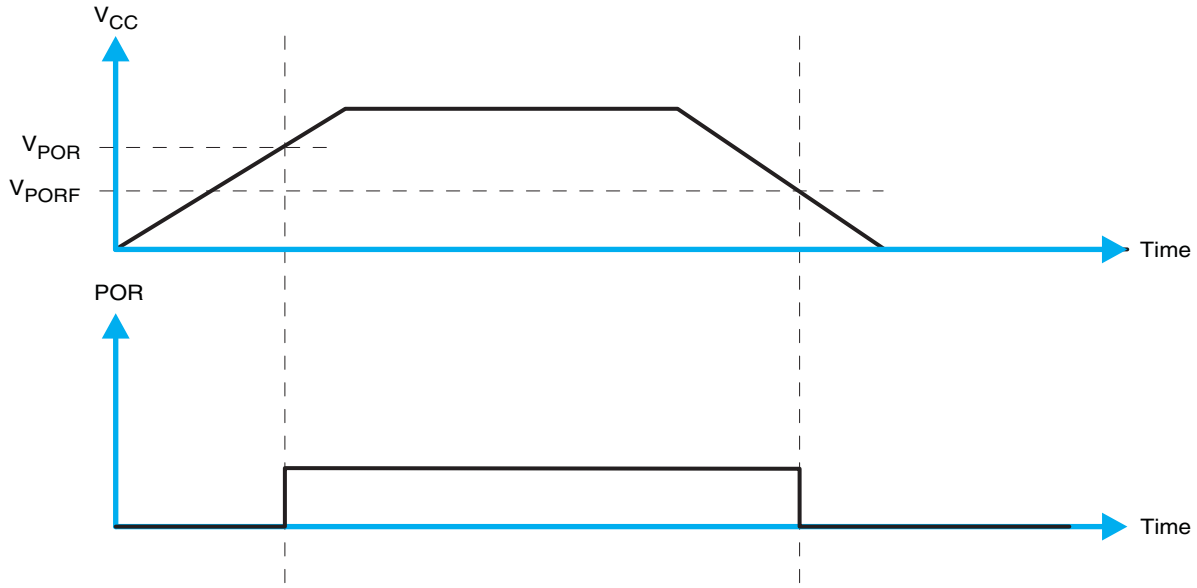


Figure 26. V_{POR}

11 Layout

11.1 Layout Guidelines

For printed circuit board (PCB) layout of the PCF8574 device, common PCB layout practices should be followed but additional concerns related to high-speed data transfer such as matched impedances and differential pairs are not a concern for I2C signal speeds.

In all PCB layouts, it is a best practice to avoid right angles in signal traces, to fan out signal traces away from each other upon leaving the vicinity of an integrated circuit (IC), and to use thicker trace widths to carry higher amounts of current that commonly pass through power and ground traces. By-pass and de-coupling capacitors are commonly used to control the voltage on the VCC pin, using a larger capacitor to provide additional power in the event of a short power supply glitch and a smaller capacitor to filter out high-frequency ripple. These capacitors should be placed as close to the PCF8574 device as possible. These best practices are shown in Figure 27.

For the layout example provided in Figure 27, it would be possible to fabricate a PCB with only 2 layers by using the top layer for signal routing and the bottom layer as a split plane for power (VCC) and ground (GND). However, a 4 layer board is preferable for boards with higher density signal routing. On a 4 layer PCB, it is common to route signals on the top and bottom layer, dedicate one internal layer to a ground plane, and dedicate the other internal layer to a power plane. In a board layout using planes or split planes for power and ground, vias are placed directly next to the surface mount component pad which needs to attach to VCC or GND and the via is connected electrically to the internal layer or the other side of the board. Vias are also used when a signal trace needs to be routed to the opposite side of the board, but this technique is not demonstrated in Figure 27.

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11.2 Layout Example

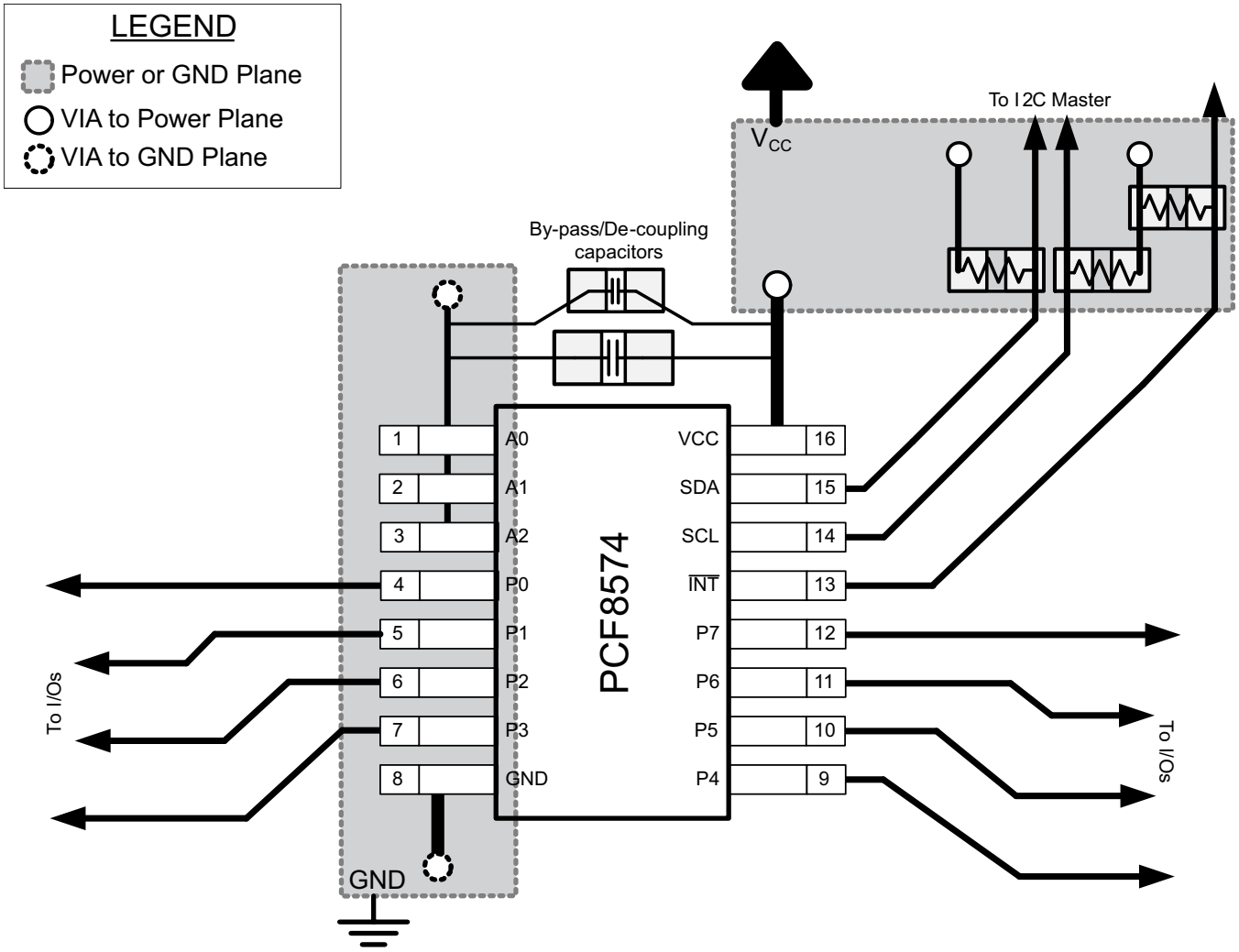


Figure 27. Layout Example for PCF8574

12 Device and Documentation Support

12.1 Trademarks

All trademarks are the property of their respective owners.

12.2 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.3 Glossary

SLYZ022 — *TI Glossary.*

This glossary lists and explains terms, acronyms and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
PCF8574DGVR	ACTIVE	TVSOP	DGV	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574DGVRG4	ACTIVE	TVSOP	DGV	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574DW	ACTIVE	SOIC	DW	16	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PCF8574	Samples
PCF8574DWR	ACTIVE	SOIC	DW	16	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PCF8574	Samples
PCF8574DWRE4	ACTIVE	SOIC	DW	16	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PCF8574	Samples
PCF8574DWRG4	ACTIVE	SOIC	DW	16	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PCF8574	Samples
PCF8574N	ACTIVE	PDIP	N	16	25	Green (RoHS & no Sb/Br)	CU NIPDAU	N / A for Pkg Type	-40 to 85	PCF8574N	Samples
PCF8574NE4	ACTIVE	PDIP	N	16	25	Green (RoHS & no Sb/Br)	CU NIPDAU	N / A for Pkg Type	-40 to 85	PCF8574N	Samples
PCF8574PW	ACTIVE	TSSOP	PW	20	70	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574PWG4	ACTIVE	TSSOP	PW	20	70	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574PWR	ACTIVE	TSSOP	PW	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574PWRE4	ACTIVE	TSSOP	PW	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	PF574	Samples
PCF8574RGTR	ACTIVE	VQFN	RGT	16	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM	-40 to 85	ZWJ	Samples
PCF8574RGYR	ACTIVE	VQFN	RGY	20	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PF574	Samples

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) RoHS: TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".
RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.
Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

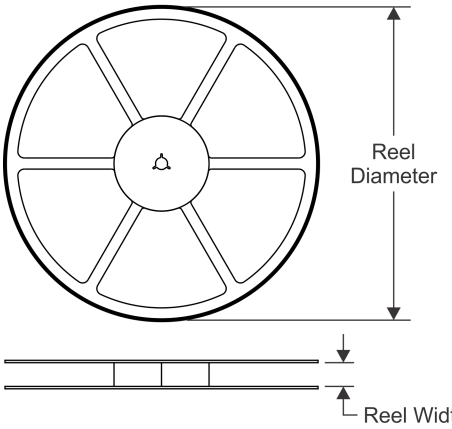
(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

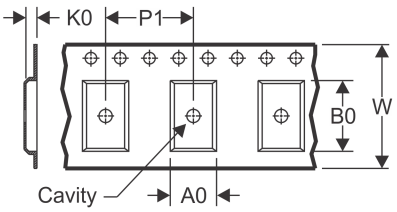
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

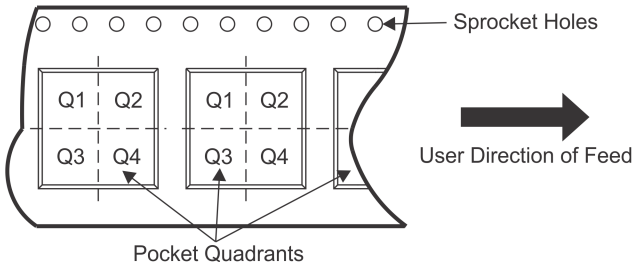


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

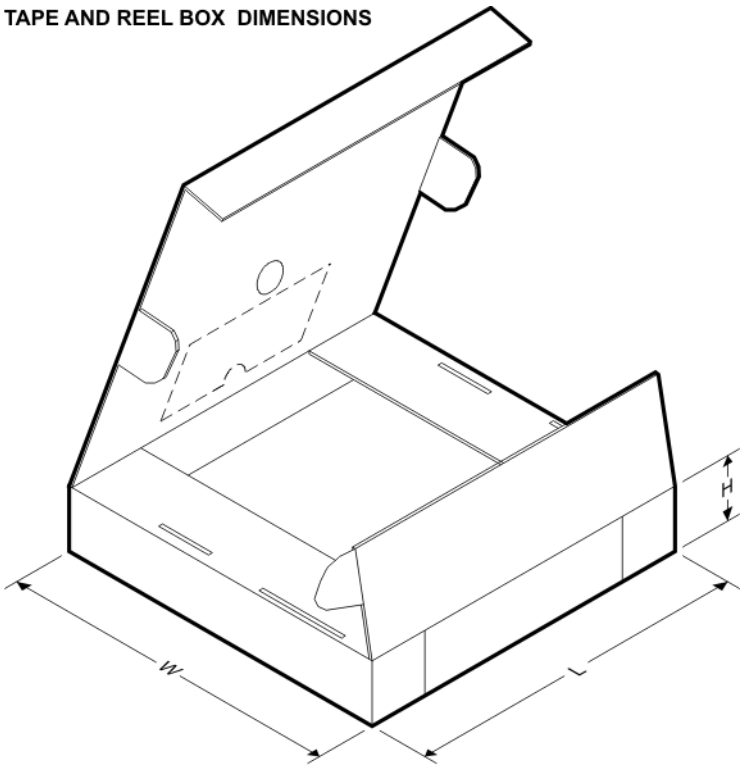
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
PCF8574DGVR	TVSOP	DGV	20	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
PCF8574DWR	SOIC	DW	16	2000	330.0	16.4	10.75	10.7	2.7	12.0	16.0	Q1
PCF8574DWRG4	SOIC	DW	16	2000	330.0	16.4	10.75	10.7	2.7	12.0	16.0	Q1
PCF8574PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
PCF8574RGTR	VQFN	RGT	16	3000	330.0	12.4	3.3	3.3	1.0	8.0	12.0	Q2
PCF8574RGYR	VQFN	RGY	20	3000	330.0	12.4	3.8	4.8	1.6	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

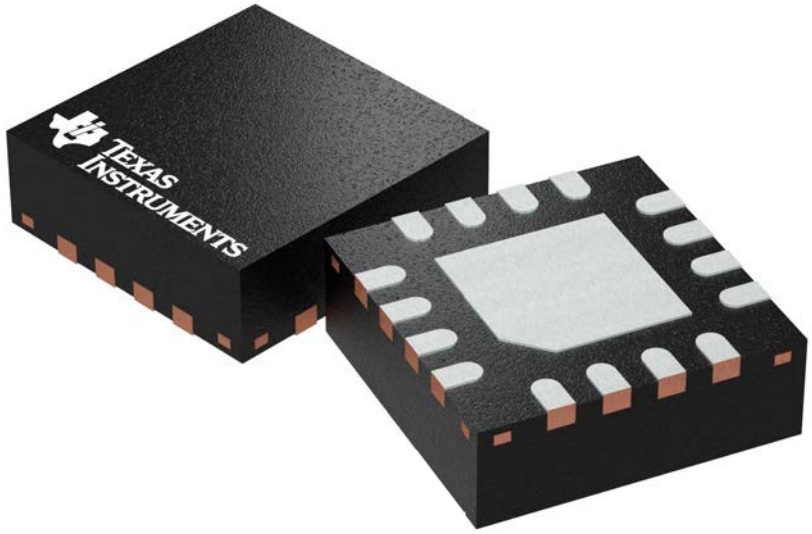
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
PCF8574DGVR	TVSOP	DGV	20	2000	367.0	367.0	35.0
PCF8574DWR	SOIC	DW	16	2000	350.0	350.0	43.0
PCF8574DWRG4	SOIC	DW	16	2000	350.0	350.0	43.0
PCF8574PWR	TSSOP	PW	20	2000	367.0	367.0	38.0
PCF8574RGTR	VQFN	RGT	16	3000	346.0	346.0	35.0
PCF8574RGYR	VQFN	RGY	20	3000	367.0	367.0	35.0

GENERIC PACKAGE VIEW

RGT 16

VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

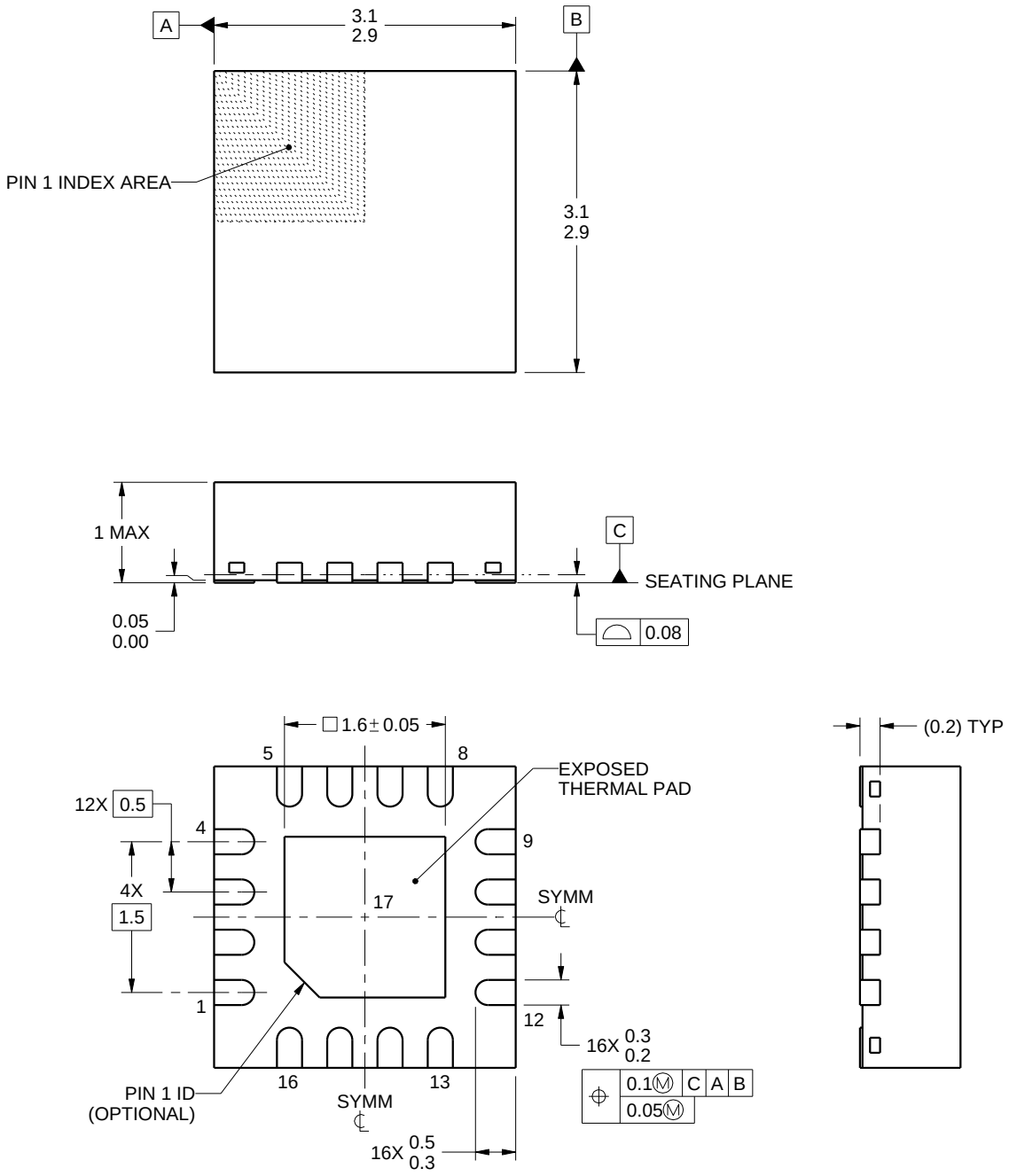
4203495/I

PACKAGE OUTLINE

RGT0016B

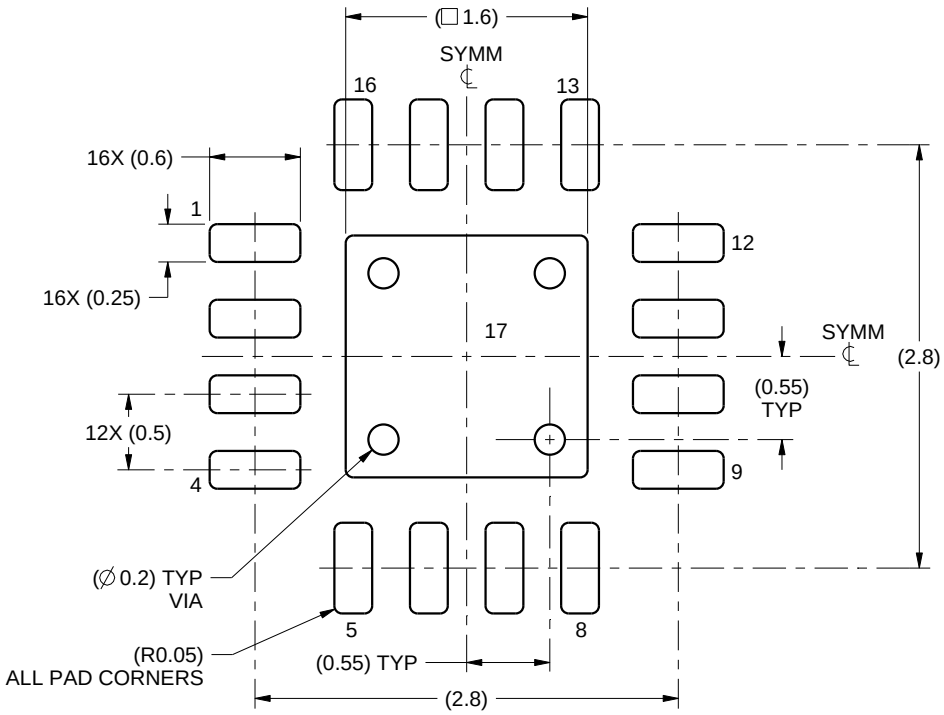
VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD

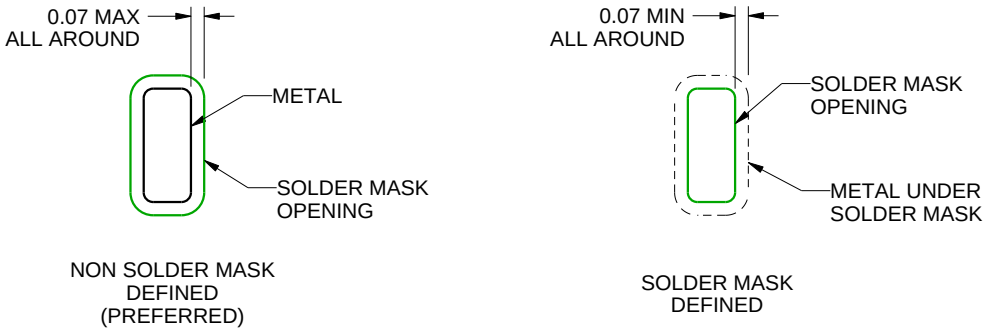


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.



LAND PATTERN EXAMPLE
SCALE:20X

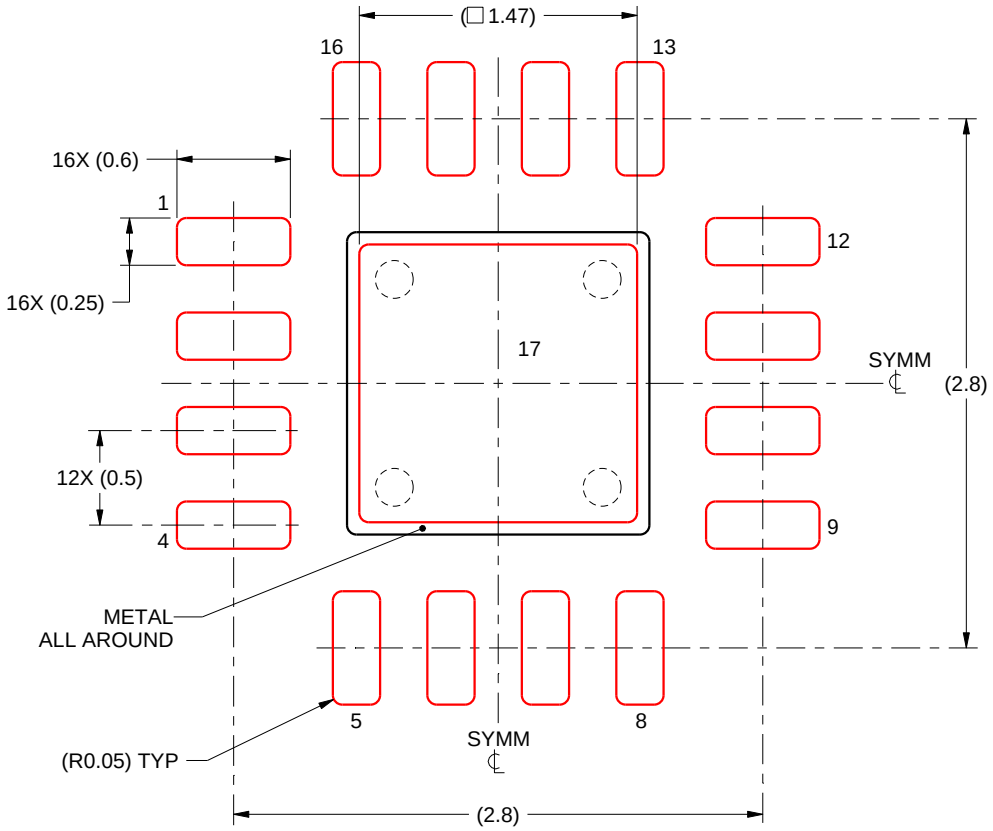


SOLDER MASK DETAILS

4219033/A 08/2016

NOTES: (continued)

4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/slua271).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL

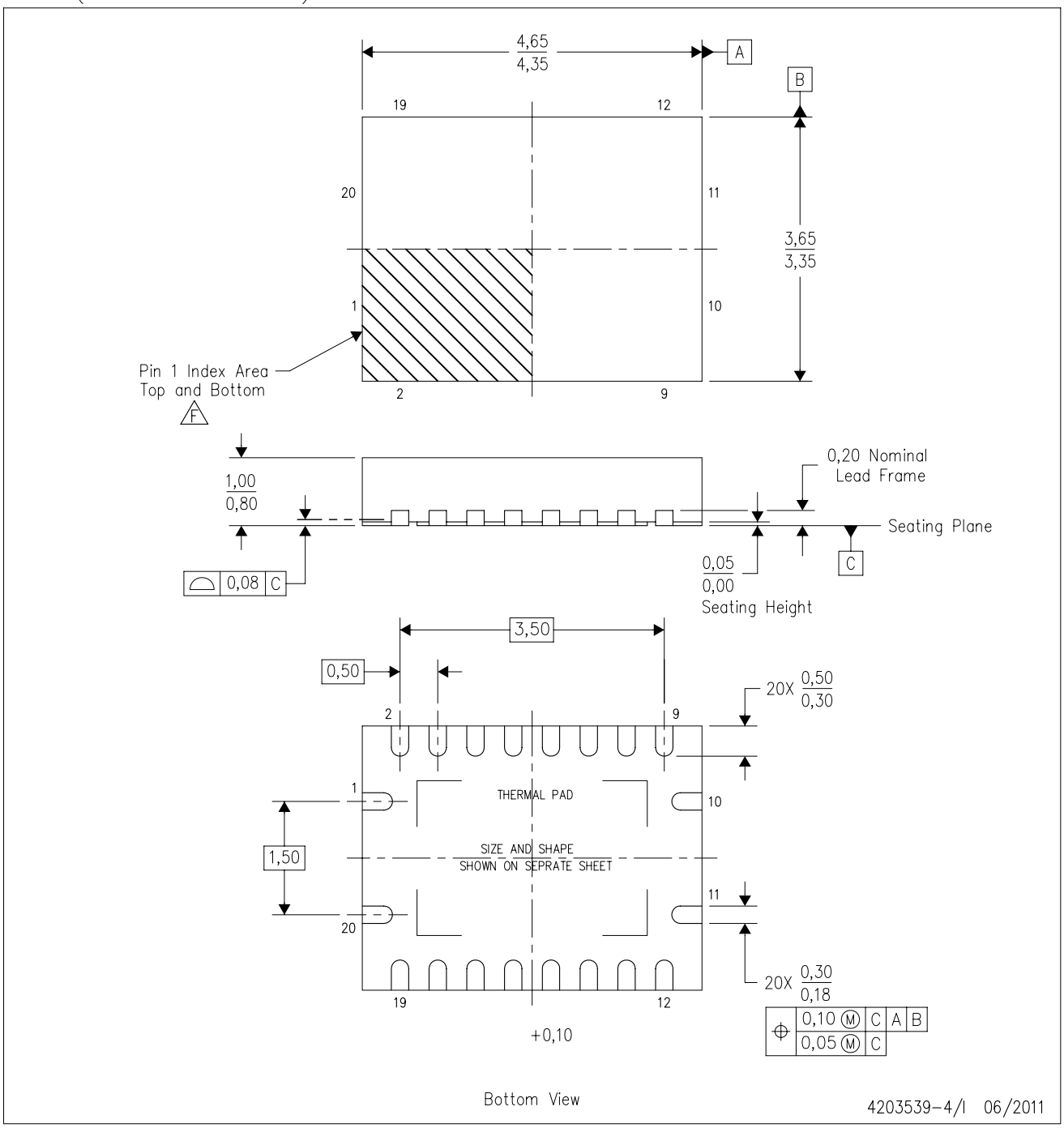
EXPOSED PAD 17:
84% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE
SCALE:25X

4219033/A 08/2016

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

RGY (R–PVQFN–N20) PLASTIC QUAD FLATPACK NO–LEAD



NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. QFN (Quad Flatpack No–Lead) package configuration.
D. The package thermal pad must be soldered to the board for thermal and mechanical performance.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
F. Pin 1 identifiers are located on both top and bottom of the package and within the zone indicated. The Pin 1 identifiers are either a molded, marked, or metal feature.
G. Package complies to JEDEC MO–241 variation BA.

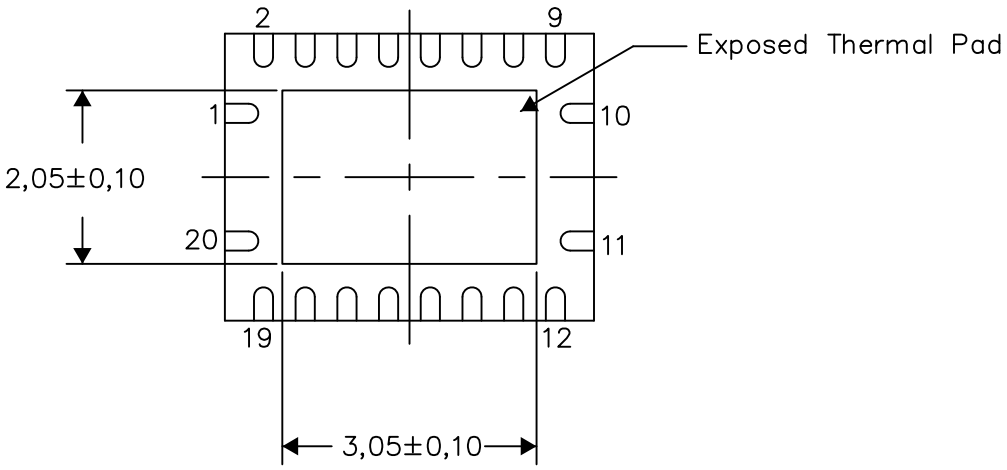
RGY (R–PVQFN–N20) PLASTIC QUAD FLATPACK NO–LEAD

THERMAL INFORMATION

This package incorporates an exposed thermal pad that is designed to be attached directly to an external heatsink. The thermal pad must be soldered directly to the printed circuit board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For information on the Quad Flatpack No–Lead (QFN) package and its advantages, refer to Application Report, QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271. This document is available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



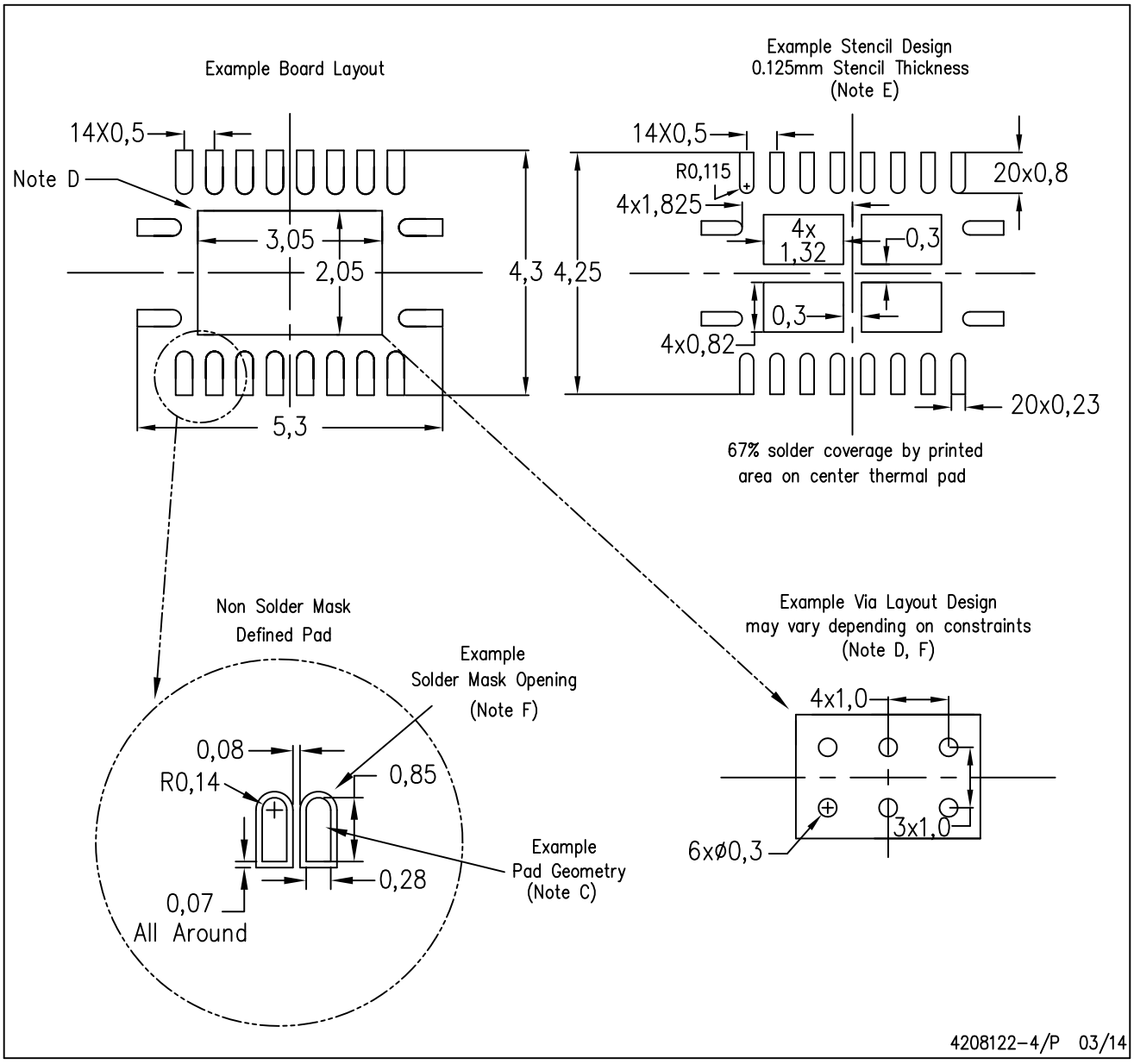
Bottom View

Exposed Thermal Pad Dimensions

4206353–4/P 03/14

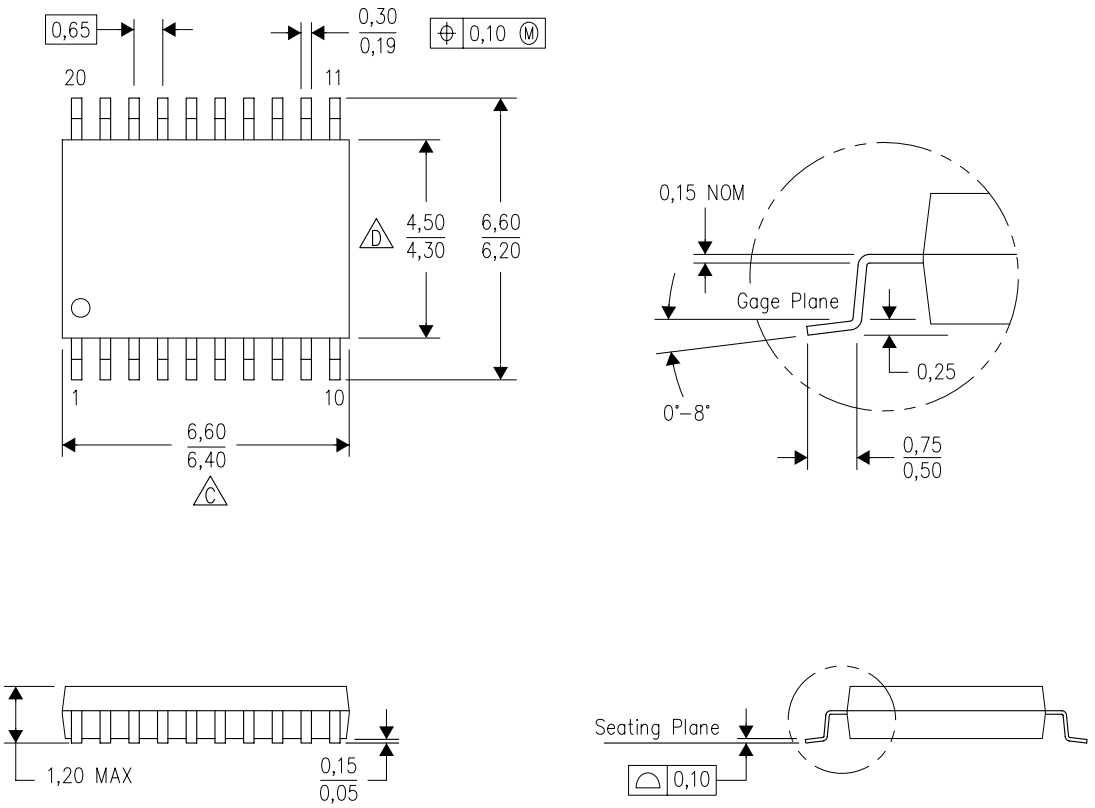
NOTE: All linear dimensions are in millimeters

RGY (R–PVQFN–N20) PLASTIC QUAD FLATPACK NO–LEAD



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Application Note, Quad Flat–Pack QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http: //www.ti.com>.
E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC 7525 for stencil design considerations.
F. Customers should contact their board fabrication site for minimum solder mask web tolerances between signal pads.

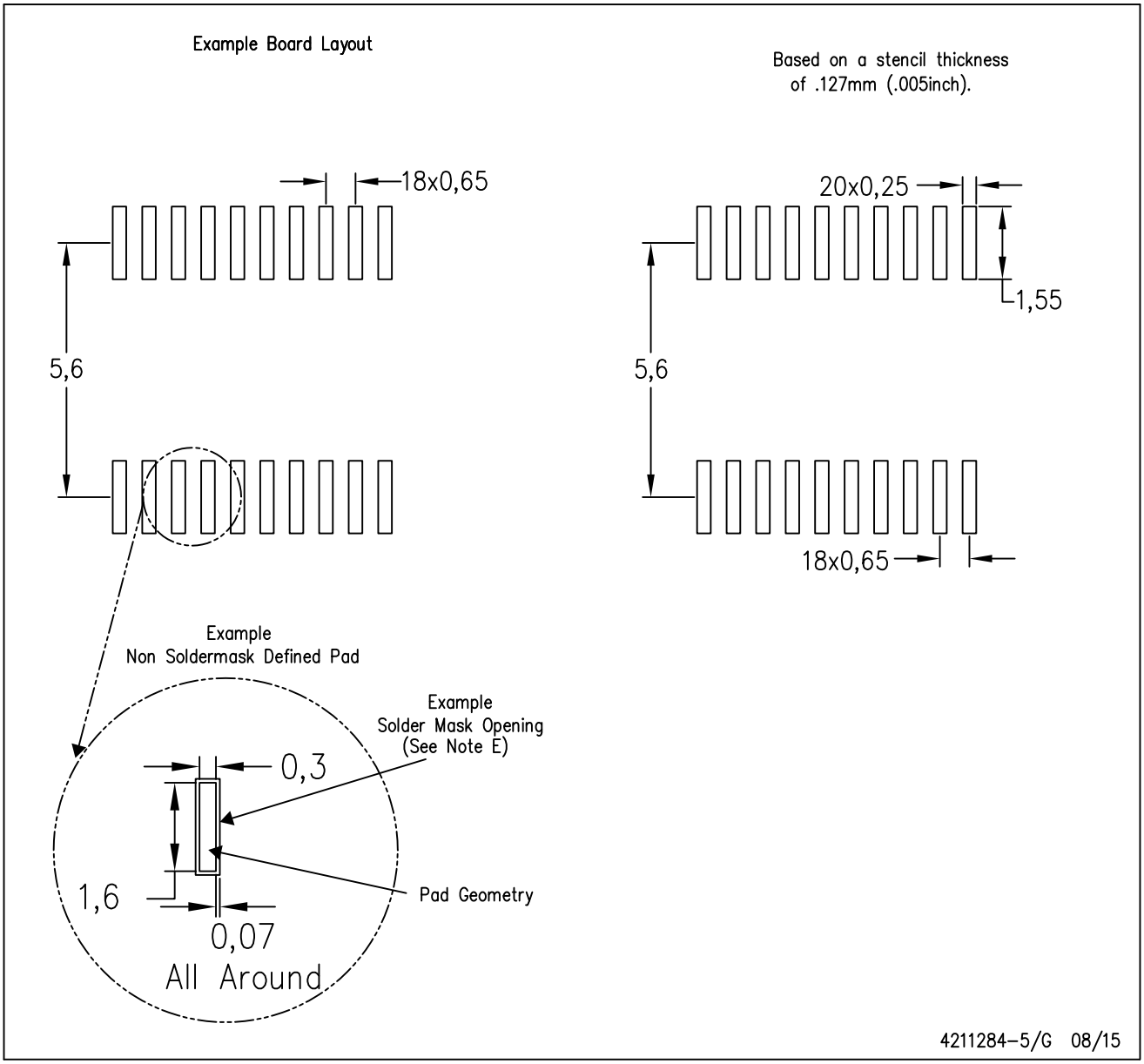
PW (R–PDSO–G20) PLASTIC SMALL OUTLINE



4040064–5/G 02/11

NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
E. Falls within JEDEC MO–153

PW (R–PDSO–G20) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate design.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

GENERIC PACKAGE VIEW

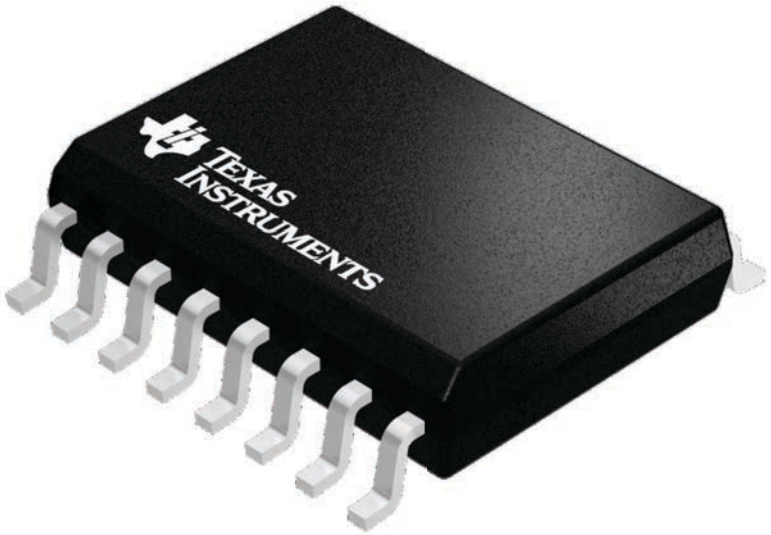
DW 16

SOIC - 2.65 mm max height

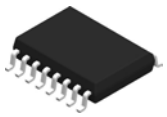
7.5 x 10.3, 1.27 mm pitch

SMALL OUTLINE INTEGRATED CIRCUIT

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4224780/A

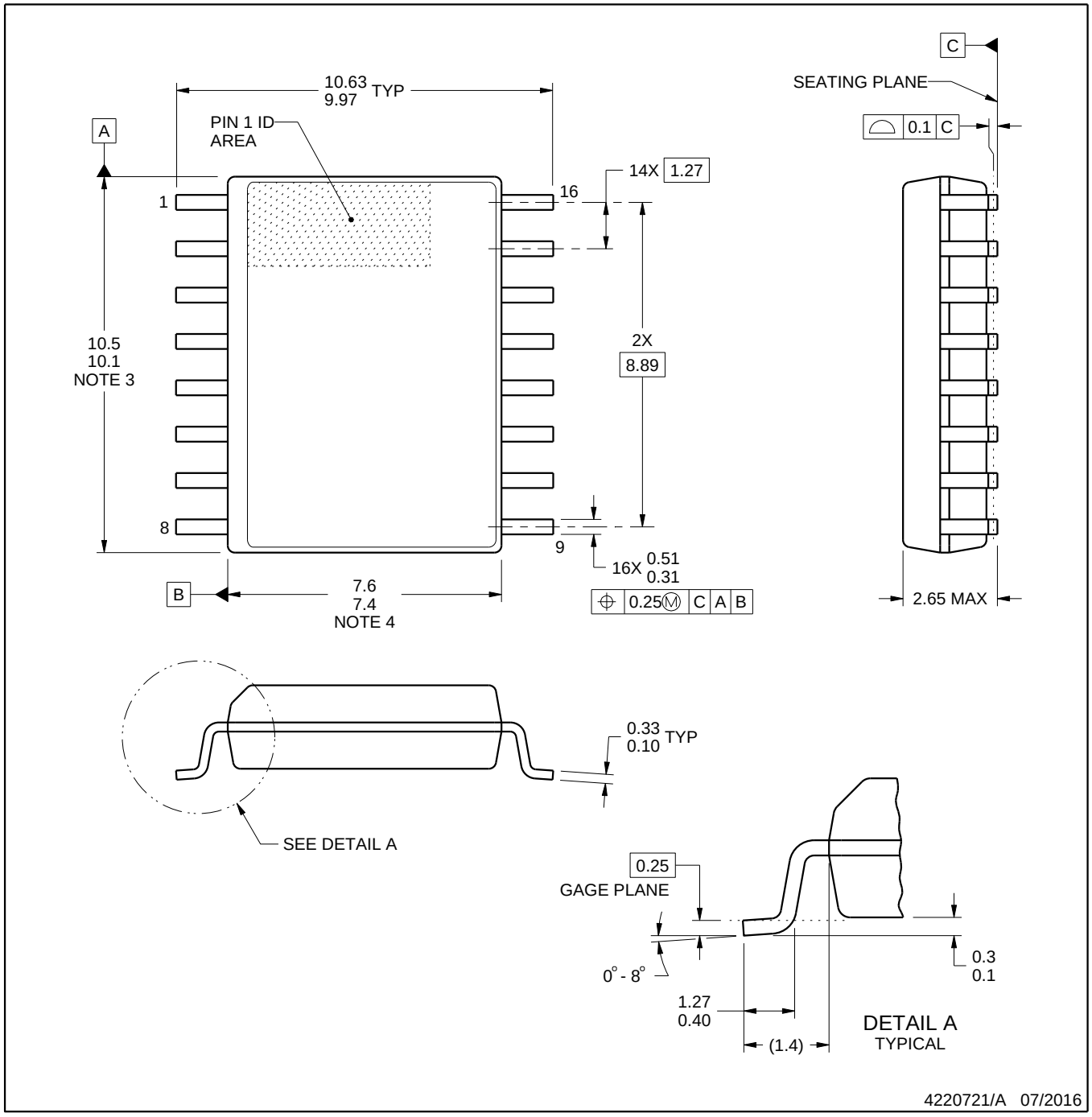


PACKAGE OUTLINE

DW0016A

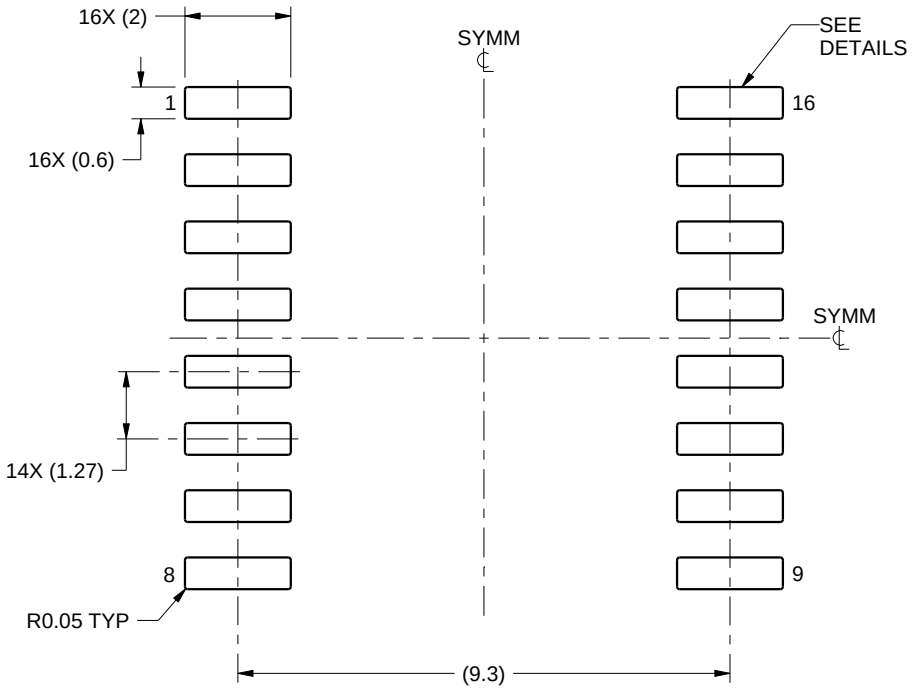
SOIC - 2.65 mm max height

SOIC

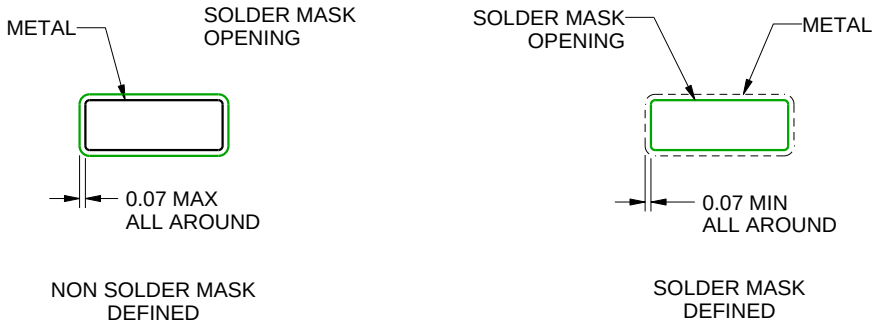


NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.
5. Reference JEDEC registration MS-013.



LAND PATTERN EXAMPLE
SCALE:7X

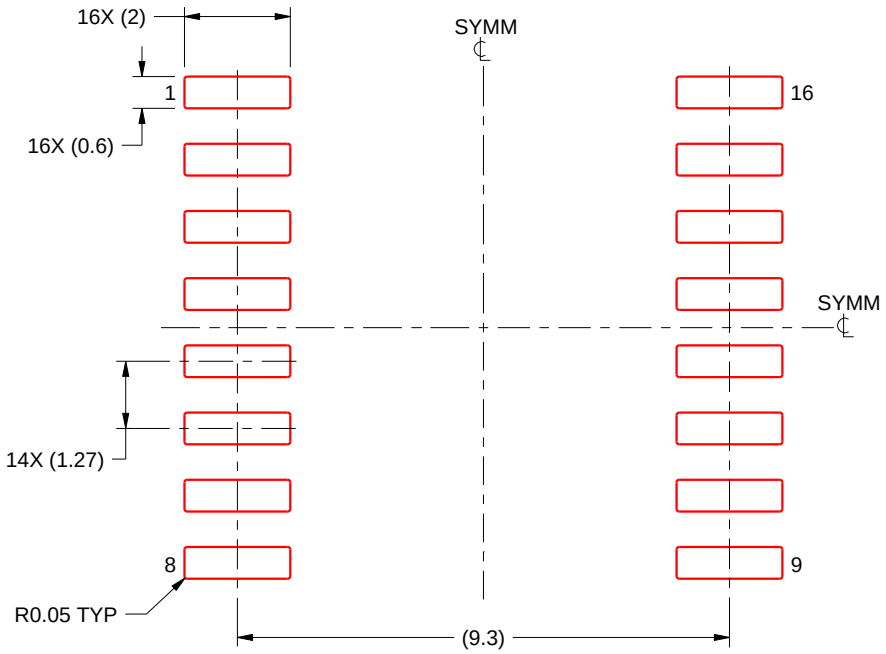


SOLDER MASK DETAILS

4220721/A 07/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:7X

4220721/A 07/2016

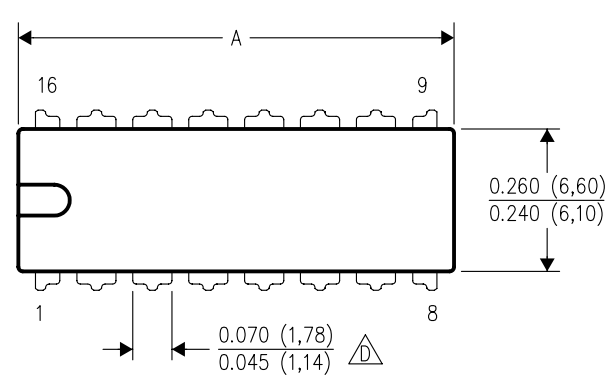
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

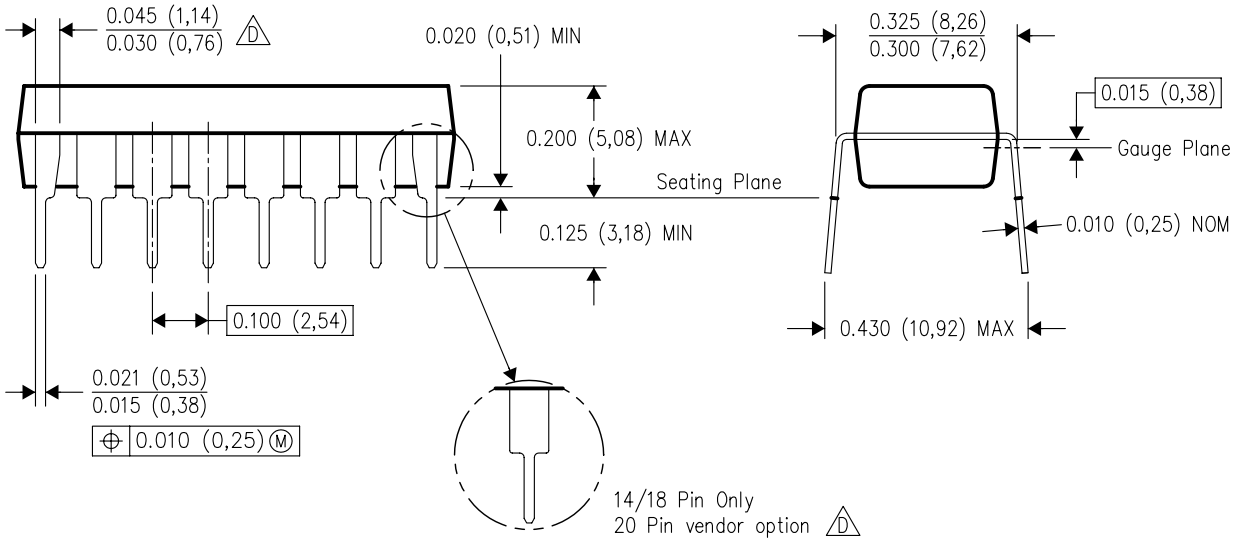
N (R–PDIP–T**)

PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN



DIM \ PINS **	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS–001 VARIATION (C)	AA	BB	AC	AD



4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

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